# The University of Sydney

## Faculty of Engineering

# AI Implementation in Healthcare

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# AI-Driven Clinical Strategy for Northern Beaches Hospital

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## I. Introduction

Northern Beaches Hospital (NBH) is a 488-bed public-private partnership (PPP) facility that opened in 2018, replacing two smaller local hospitals. Despite its modern infrastructure, NBH has faced significant challenges in providing high-quality safe care to public patients(1; 2). A 2025 NSW Auditor General report found that NBH was not effectively delivering the best clinical outcomes for the community(3). These issues were thrust into public view by a tragic incident: the death of two-year-old Joe Massa in September 2024 after a severe care delay in NBH's emergency department (ED)(4). An investigation revealed he was mis-triaged as a less-urgent category and waited over 2.5 hours for treatment despite critical vital signs(2). This case spurred intense scrutiny of NBH's operations, with the state government introducing "Joe's Law" to ban future hospital PPPs(5). Multiple root causes underlie NBH's failures: overburdened personnel, process breakdowns, outdated systems, and misaligned organisational structure. NBH's electronic health record (EHR) system had known safety risks since 2018 but remained unaddressed(6).

Advanced data-driven and AI solutions offer a promising pathway to address these issues. These AI solutions are being recognised in their ability to supplement clinical decision making and general workflow. In a healthcare setting especially, AI tools may serve to reduce the significant time spent on documentation.

This report outlines an AI implementation plan designed for the benefit of the NBH ED. This report conducts initial analysis into the context of the NBH ED before contacting several stakeholders of the organisation to obtain specific insights as to the day-to-day functionalities and difficulties associated with the processes and systems currently employed at the hospital. Through these insights, we found a number of AI technologies suited to healthcare implementation and analysed these against stakeholder priorities to ultimately identify an AI system suited to the NBH ED. Finally, we reflect on this decision by highlighting associated risks and any challenges which may arise from implementation, before providing short and long-term recommendations based upon all previous information.

## II. Research Methodology

### A. Stakeholder Engagement and Data Collection

To ensure the solution addresses real needs, we engaged some stakeholders through semi-structured interviews to gather insights on current workflows, pain points, and ideas for improvement. The interviewees described daily experiences in the ER, highlighting challenges like excessive time spent on EHR data entry, interruptions to patient care for documentation, and the impact on staff morale. They also discussed attitudes toward potential technological solutions (including voice recognition tools) and any concerns or expectations. The interview was recorded with consent, producing qualitative data ($\sim$60 minutes of dialogue) for analysis. The goal was to ground our problem definition in first-hand stakeholder evidence and ensure any proposed AI intervention aligns with clinician's needs and organizational context.

### B. Thematic Analysis of Stakeholder Input

We then analysed interview transcript using thematic analysis (an inductive qualitative method) to identify recurring themes and concerns. We followed a process: familiarisation with the transcript, categorising (open coding) significant statements, clustering of codes into candidate themes, and iterative refinement of those themes. Initial coding yielded numerous keywords (e.g. "documentation," "waiting," "burnout," "technology"), which were then grouped by conceptual similarity. Table 1 below summarizes the key themes that emerged, as well as a brief description and illustrative quotes from the stakeholder:

From the thematic analysis, it's clear that documentation burden came up as the most prominent issue. The stakeholder repeatedly expressed frustration about how much time doctors spend updating patient charts — often identifying that such copious documentation is at a risk of harming patient care standards ("how much time do you want to spend on the computer? Like, I want to actually spend time with the patients?"). This not only slows down the workflow but also contributes to patient backlogs, as clinicians are constantly balancing paperwork with seeing the next patient. This sentiment reflects broader findings across the industry, where doctors reportedly spend up to two hours on paperwork for every hour of patient

TABLE I: Thematic Analysis Results (Stakeholder Interview)

| Theme | Description | Example Stakeholder Comment |
|---|---|---|
| Documentation Burden | Time-consuming charting tasks that take clinicians away from patients. | "How much time do you want to spend on the computer? Like, I want to actually spend time with the patients?" |
| Workflow Inefficiencies | Delays and bottlenecks in ER processes, often linked to admin tasks. | "I would say interruption is, one of the describing words of the job" |
| Technology & EHR Issues | Suboptimal existing IT systems (EHR usability, lack of integration). | "I think the most frustrating thing is that it's not all in one place. It's all in different places" |
| Clinician Well-being | Impact of work conditions on staff fatigue and burnout. | "we don't we don't discuss our workload. We just work late" |
| AI Caution | The necessity for exhibiting caution in integrating AI tools into the healthcare system | "Where we're at at the moment with AI would be more to assist in documentation or doing manual tasks" |

care (7). Concerns about current tech also came up: the hospital's EHR system was described as clunky and time-consuming, with no voice input support, only adding to the delay. Clinician well-being was another major concern, with comments pointing to fatigue and burnout caused by excessive administrative tasks. One such stakeholder reflected on the reality of long working hours for the teams of Doctors in hospitals in saying "we don't we don't discuss our workload. We just work late". Considerably, the patient experience also suffers; patients can tell when their doctor is focused more on the screen than on them, which can hurt trust and satisfaction.

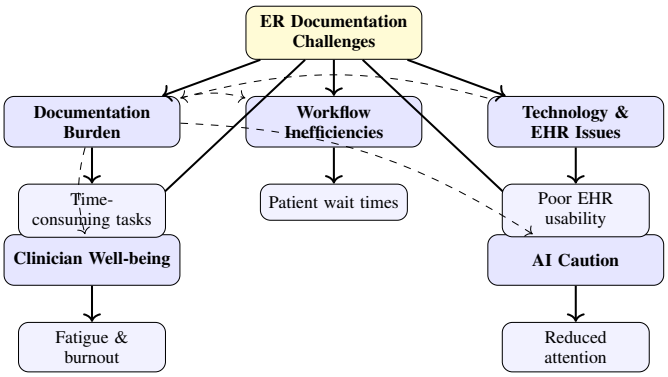


Fig. 1: Thematic coding of stakeholder interview data revealing key themes and subthemes.

### C. Pattern and Sentiment Analysis

Beyond identifying themes, we analysed patterns and sentiment in the stakeholder's feedback. This involved noting the frequency/emphasis of each theme and the emotional tone (positive, negative, neutral) expressed. The stakeholder's sentiments help clarify which problems are most urgent and how receptive they might be to new solutions:

As shown in Table 2, the overall sentiment around the current state of ER documentation and workflows is strongly negative. The stakeholder is frustrated and concerned, particularly about documentation burden and its cascading effects. Importantly, however, there is a glimmer of positive sentiment in discussing potential solutions – the interviewee seemed cautiously optimistic that an AI-driven transcription tool could alleviate some problems ("What would be useful is to have an AI scribe. That would be amazing"). This mix of frustration with the status quo and openness to innovation informed our problem definition and ideation. The data suggested that an effective solution should primarily target the automation of simple patient documentation to free up clinician time, thereby improving workflow efficiency and reducing burnout. The idea of an AI scribe system naturally emerged from these insights, aligning with the stakeholder's expressed needs and hopes.

Throughout the analysis, we cross-checked emerging themes with raw data to ensure trustworthiness (e.g., using direct quotes as evidence). The thematic and sentiment analysis provided a clear mandate: any proposed intervention should significantly reduce manual documentation in the ER. This became the focal problem statement that guided brainstorming in the ideation and problem definition phase and ultimately led to our specific choices for our AI system.

### D. Stakeholder Acceptance Factors

Based on sentiment analysis, clinicians are open to AI assistance if it directly reduces documentation time, improves work-life balance, and is non-intrusive. Key factors affecting acceptance include:

- **Transparency**: Staff must understand how AI works and that final note control remains with them.
- **Training and Involvement**: Involving clinicians from the pilot stage and offering role-specific onboarding which could help foster early trust.
- **Ethical Considerations**: Informing patients of ambient listening, providing opt-out choices, and safeguarding privacy must be core to implementation.

Change management strategies used in similar deployments (e.g., UM Health-West) emphasise clinician wellness and reduce fears of job displacement(19).

### E. Ideation and Solution Selection Process

Our solution ideation process began with a divergent phase generating 12 potential AI interventions across the four

TABLE II: Pattern & Sentiment Analysis of Stakeholder Feedback

| Theme | Sentiment (tone of feedback) | Observations / Pattern |
|---|---|---|
| Documentation Burden | Negative (Frustrated) – Strong pain point. | Mentioned frequently and emphatically as a major issue affecting daily work. |
| Workflow Inefficiencies | Negative (Concerned) – Noted as problematic. | Attributed primarily to documentation delays; directly linked to patient wait times. |
| Technology & EHR | Mixed – Frustration with current state, hope for improvement. | Criticized current EHR; expressed interest in better tools (e.g. "either a camera or a microphone recording what was said, that would be a huge time saver"). |
| Clinician Well-being | Negative (Burnout risk) – Described strain. | Implicitly tied to documentation load; stakeholder conveyed worry about burnout if nothing changes. |
| AI Caution | Hesitant – Nervous about AI overreaching in its clinical capacity. | Acknowledged that AI tools are not advanced and trusted enough to offer clinical decision-making support. |

TABLE III: Pattern and Sentiment Analysis of Key Themes

| Theme | Cross-Stakeholder Patterns | Sentiment Analysis | Underlying Systemic Factors | Attempts to Address |
|---|---|---|---|---|
| Documentation Burden | Widely raised by clinicians and admin | Strong frustration, urgency | Time-consuming EHR use | After-hours charting |
| Workflow Inefficiencies | Noted across clinicians and staff | Concern, moderate urgency | Delayed discharge processes | Triage/queuing adjustments |
| Technology Limitations | Acknowledged by clinicians | Mixed: hopeful and critical | Lack of integration with voice tools | Basic templates and manual entry |
| Burnout and Wellbeing | Focus of clinician concerns | Emotional intensity, fatigue | Long shifts, repetitive tasks | Rotas, wellness initiatives |
| Patient Communication | Highlighted by patients/nurses | Regretful, low engagement | Split attention during consults | Communication training |

themes. These candidates underwent a three-stage evaluation process:

First, a **capability-need mapping** assessed each solution's relevance to identified problems (Table VI). Second, we conducted a **feasibility assessment** using a weighted criteria matrix evaluating technical maturity, implementation complexity, cost, and stakeholder acceptance (Table VII, see Appendix). Finally, we performed a **systems impact analysis** to identify potential synergies between solutions.

This systematic evaluation narrowed our focus to three high-potential AI domains: (1) scribe for patient visits, (2) documentation automation, and (3) patient risk monitoring. Through extensive refinement and conversation with stakeholders, we have arrived at a consensus of which AI tool would suit the contextual requirements of NBH ED, whilst remaining relatively uncomplicated in its ethical and legal integration.

## III. AI System Proposal: Nuance Dragon-Style Scribe

Based on the stakeholder-driven problem definition, the chosen solution is an AI-powered scribe system for the ER, modeled on Nuance's Dragon Ambient eXperience (DAX) technology. This system will serve as a "virtual medical scribe," automatically transcribing doctor–patient conversations and generating structured clinical documentation in real time. The core components of the proposed system include speech recognition, natural language processing (NLP) for medical language, and seamless integration with the hospital's EHR. The solution leverages state-of-the-art NLP (potentially large language models like GPT-based technology) to ensure the transcribed notes are comprehensive and contextually accurate(8).

**How it works:** During an ER encounter, the physician wears a microphone or uses a dedicated mobile app that securely captures the dialogue between clinician and patient. The audio stream is processed by a speech-to-text engine (Nuance Dragon Medical's speech recognizer) which is pretrained on medical vocabulary and can transcribe clinical speech with high accuracy (up to $\sim$99% under ideal conditions(9)). The transcribed text is then fed into an NLP module that interprets the content, identifying key clinical information (e.g. symptoms, history, diagnostics, treatment plans). Using advanced algorithms (including large language models), the system generates a structured clinical note summarising the encounter in a format suitable for the EHR (e.g. SOAP note sections). This draft note typically includes the chief complaint, history of present illness, review of systems, physical exam findings (that were verbalized), impressions, and plan.

The Nuance Dragon-based scribe operates in near real-time: by the end of the patient visit, a full draft of the encounter note is ready for review. The clinician then reviews and edits

the note for any needed corrections or additions (e.g. adding a detail from a physical exam that wasn't spoken aloud) before finalising it in the EHR. This human verification step is critical to ensure accuracy and completeness(10) – the physician remains the ultimate authoriser of the documentation. Once approved, the note is saved to the EHR just as if the clinician had typed it, and can be forwarded or used for discharge summaries, prescriptions, billing codes, etc.

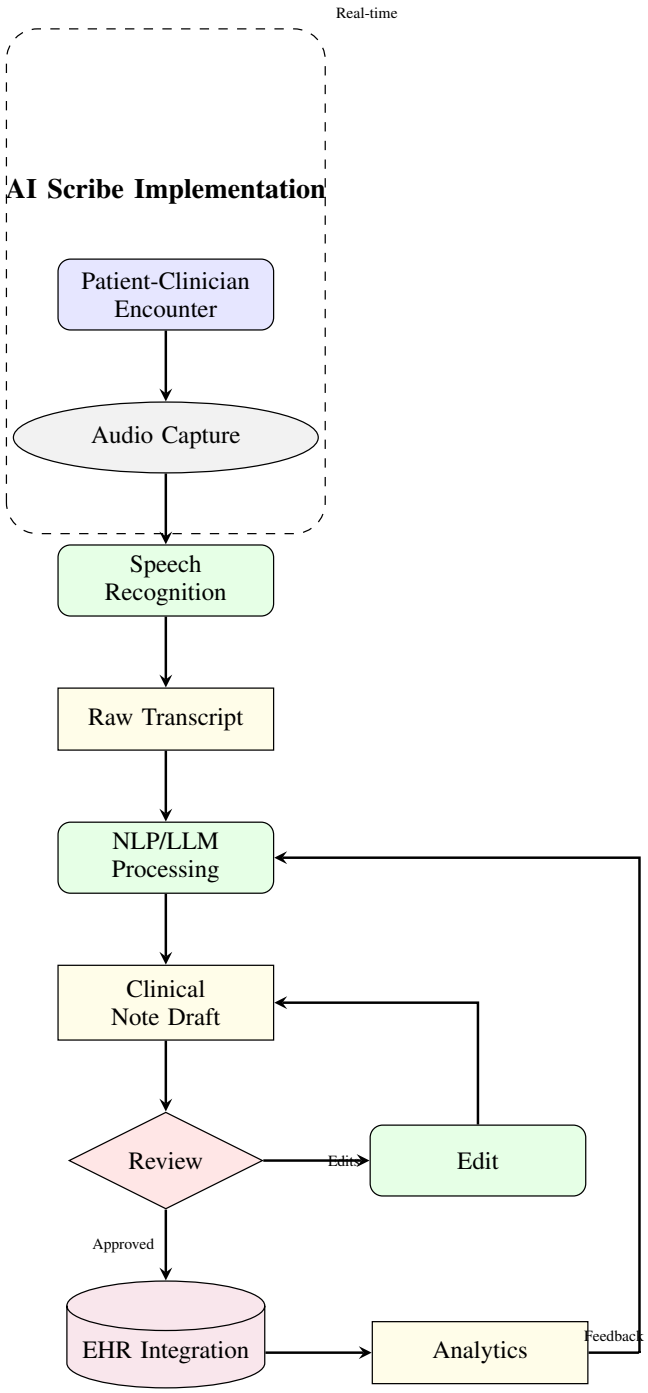


Fig. 2: AI Scribe System Implementation flowchart showing data flow from patient encounter through EHR integration.

### A. *Benefits to ER efficiency and care:*

The AI scribe system directly addresses the major themes identified by the ER stakeholder. By offloading the documentation task to an automated assistant, clinicians can spend less time typing and more time with patients. This has multiple cascading benefits:

- **Time Savings:** Nuance DAX can save doctors an average of 7 minutes per encounter and reduce total documentation time by 50%(11). In fast-paced ER settings, this adds up — allowing clinicians to see more patients or finish charting within their shift. Beacon Health System reported that Dragon Medical + DAX enabled doctors to see 5–7 more patients per week(12).
- **Improved Workflow & Throughput:** Real-time documentation during consultations speeds up patient flow and reduces wait times(13). It also prevents charting delays from holding up admissions, discharges, or transfers.
- **Enhanced Accuracy and Quality:** The AI scribe captures the full conversation, reducing omissions and improving note completeness(14). 79% of DAX users reported better documentation quality(15). Structured output also supports smoother billing and coding processes.
- **Clinician Satisfaction and Well-being:** Reducing clerical tasks helps combat burnout. Some hospitals saw burnout drop as much as if users had switched to part-time(16). Staff also report stronger engagement with patients, as they can maintain eye contact and natural conversation(11; 15). In one Nuance survey, 83% of patients felt their doctor was more focused(15).
- **Integration with Existing Systems:** Nuance Dragon integrates smoothly with major EHRs. For example, DAX Copilot works within Epic workflows(17). This ensures minimal disruption, high adoption, and compatibility with existing documentation templates.

### B. *System architecture:*

The AI scribe uses a modular system with the following components: (1) **Audio Capture:** A secure app records clinician-patient conversations in real time. (2) **Speech Recognition:** Converts speech to text using a medical-grade model. (3) **NLP/LLM Processing:** Interprets and structures the text into a clinical note (e.g., separating vitals, meds, and narrative). (4) **Output Interface:** Sends the draft note to the EHR or a review app. (5) **EHR Integration:** Uses APIs or HL7/FHIR to insert the signed note into the patient record.

The system is HIPAA-compliant, with encrypted transmission and storage. Cloud-based processing (e.g., Microsoft Azure) can be used while ensuring no raw audio is stored — only transcripts and notes are retained(14). Functionally, it mirrors a human scribe but with greater speed and consistency.

In short, the Nuance Dragon-style AI scribe directly addresses documentation inefficiencies in the ER. It improves workflow, note quality, and clinician well-being. The next section discusses risks and how to manage them. (See Appendix C for technical specs and a sample AI-generated note.)

TABLE IV: Capability-Need Mapping for AI Scribe Integration

| Thematic Need | Sub-component | Current Workarounds | Required Capabilities | Potential AI Approaches |
|---|---|---|---|---|
| Documentation Burden | Manual EHR entry | Typing during or after shift | Hands-free, ambient documentation | Nuance DAX NLP engine |
| Workflow Inefficiencies | Long patient wait times | Delayed note finalization | Real-time note generation | Speech-to-text with EHR API |
| Technology Limitations | EHR lacks voice input | Keyboard-based navigation | Seamless voice-to-text input | Dragon Medical One integration |
| Burnout and Fatigue | Admin overload | Overtime and weekend work | Reduced cognitive load | Auto-formatting of SOAP notes |
| Patient Communication | Split attention | Clinicians face screen more than patient | Passive audio capture | Ambient listening + summarization |

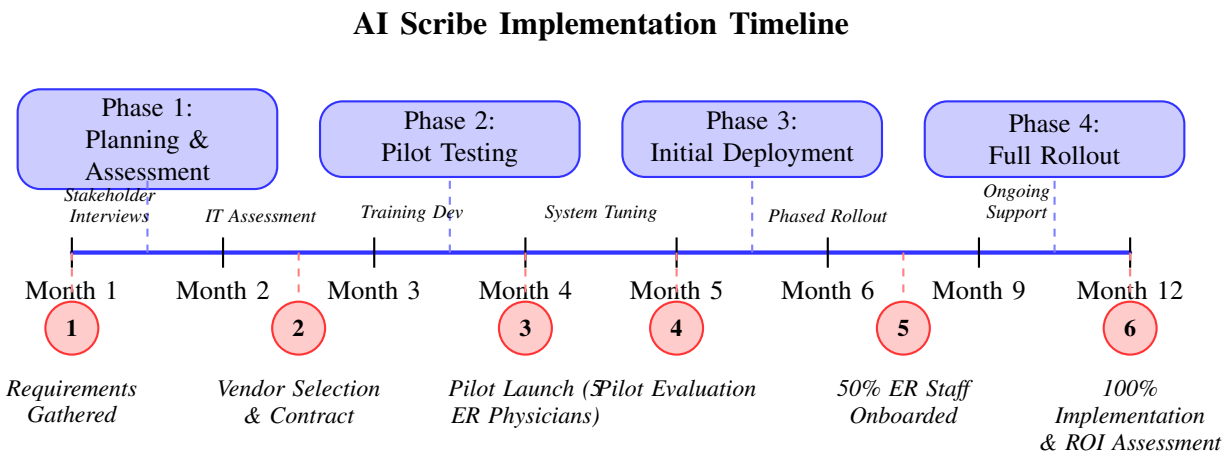


Fig. 3: Projected timeline for the AI scribe system implementation across the Emergency Department, including major phases and milestones from planning through full deployment.

## IV. Risks and Mitigation Strategies

While the implementation of an AI scribe system offers significant clinical and operational benefits, it also introduces a range of potential risks that must be carefully managed. These risks span transcription accuracy, data privacy, clinician liability, system integration, user acceptance, automation over-reliance, cost justification, and patient consent. Each of these challenges, if unaddressed, could impact safety, trust, or sustainability. However, the project's design includes comprehensive mitigation strategies that are tailored to each risk domain. For example, clinicians will remain the final authority on documentation to ensure safety and legal protection, while HIPAA-compliant infrastructure and strict access controls will protect patient privacy. To promote adoption, the system will be introduced in phases with pilot users, training, and feedback loops. Cost-related concerns are addressed through staged rollout and ongoing ROI tracking. Table V summarises these core concerns, supported by case study insights.

### A. Case Study Examples:

The experiences of other healthcare organisations with AI scribe technology reinforce both the potential benefits and the importance of mitigation strategies:

**Beacon Health System (Indiana, USA):** Beacon implemented Nuance's Dragon Medical One and DAX across several clinics to address documentation delays(12; 20). They reported a 25% reduction in time spent per patient consultation and were able to reduce reliance on human scribes by 50% in some departments(12). Clinicians noted they could focus on patients without worrying about writing notes, echoing improvements in patient engagement. Beacon's success was attributed to strong leadership support and training – clinicians were guided on how to incorporate DAX into their routine, and initial skepticism was addressed by demonstrating that note quality remained high. This case illustrates that efficiency gains are real, but achieving them requires user buy-in and workflow adjustments, which Beacon managed through proactive change management.

**University of Michigan Health-West:** This health system focused on reducing burnout by using DAX in primary care (their findings are applicable to ER as well)(19; 19). After a year, clinicians using DAX >60% of the time saw significantly less burnout and could see about 1 extra patient every two clinic days on average(16). Notably, UMHW's

TABLE V: AI Scribe Implementation Risks & Mitigation Strategies

| Risk | Description | Mitigation Strategy |
|---|---|---|
| 1. Transcription Errors | Misinterpretation of speech, clinical terms, or context could result in incomplete or inaccurate notes. | Mandatory **clinician review** of AI-generated notes. Ongoing **model refinement** using feedback. Start with **pilot phase** to test accuracy. Flag low-confidence transcriptions. |
| 2. Data Privacy & Security | Audio and text data may expose sensitive PHI, risking breaches or regulatory noncompliance. | Use **HIPAA-compliant** and **HITRUST-certified** systems. No raw audio storage. **End-to-end encryption**. Transparent **consent process** for patients. Conduct regular security audits. |
| 3. Clinical Liability | If the AI omits critical findings or details, it may cause care errors or expose staff to legal risks. | Clinicians retain **full responsibility** for notes. Training to **verbalize findings** clearly. Maintain **textual transcripts** as audit trails. Include **legal disclaimers** clarifying AI's assistant role. |
| 4. Integration Issues | Poor interfacing with EHRs could disrupt documentation flow or lead to data loss. | Conduct **early integration testing** with EHR vendors. Use **FHIR or HL7 APIs**. Ensure **real-time syncing**. Implement **IT support** during rollout. Plan fallback mechanisms. |
| 5. User Resistance / Low Adoption | Clinicians may distrust the AI, feel uncomfortable being recorded, or fear increased workload. | Launch with **early adopters** as champions. Frame as a **wellness initiative** (not surveillance). Offer **training**, live support, and **feedback loops**. Share early success metrics. |
| 6. Over-Reliance on Automation | Clinicians may become passive or fail to catch errors, trusting the AI too much. | Emphasize that **human judgment is final**. Introduce **routine audits** of notes. Rotate **non-AI weeks** for skill retention. Encourage a **co-pilot model**, not autopilot. |
| 7. Cost & ROI Concerns | Subscription, infrastructure, and training costs may not yield measurable returns. | Begin with **limited deployment** and track ROI (e.g., time saved, patients seen). Present case studies showing **proven returns**. Adjust licensing tiers based on usage. |
| 8. Ethical & Consent Concerns | Some patients may not consent to being recorded or may feel their privacy is violated. | Provide **clear disclosure** and obtain verbal or written **consent**. Offer **opt-out options**. Involve ethics board in protocol design. Document consent in EHR. |

leadership framed the AI scribe as a wellness initiative, not just a productivity tool, which improved adoption(19). They also started with tech-enthusiastic physicians and gradually expanded, which helped work out kinks and create internal champions. The UMHW case underscores that emphasizing clinician-centric goals (like getting doctors home on time, as one physician remarked(19)) and starting small can lead to both happier staff and measurable throughput gains. It also highlights that even with clear benefits, continuous encouragement and clarity of purpose are needed to get clinicians to trust and use the system consistently.

Conclusively, while there are valid risks in deploying an AI scribe, none are insurmountable. With thorough planning – ensuring privacy, maintaining human oversight, integrating well with existing systems, and actively managing change – the ER can safely harness this technology. The experiences of early adopters show that when done right, an AI scribe can improve efficiency without compromising patient care or documentation quality(10). Our implementation plan (see Appendix E for a draft roadmap and risk register) incorporates these lessons to maximize success.

### *B. Ethical and Legal Considerations*

Our proposed AI system introduces important ethical and legal considerations that must be addressed proactively. From an **ethical standpoint**, three key themes emerged: transparency and explainability of AI decision-making, preserving clinician autonomy, and ensuring equitable performance across all patient demographics. These principles are especially critical for the triage AI component, which must avoid replicating historical biases embedded in healthcare delivery systems. A well-documented case showed that a widely used healthcare algorithm systematically underestimated illness severity in Black patients, underscoring the need for fairness audits and inclusive design(21). To support ethical governance, we propose establishing an AI Ethics Advisory Board with representatives from clinical, technical, and patient communities. A full ethical framework aligned with best-practice AI principles is detailed in **Appendix B-D**, including Table XVI, which outlines implementation mechanisms for transparency, fairness, autonomy, and accountability.

From a **legal and regulatory perspective**, several frameworks govern the deployment of this system. These include the Therapeutic Goods Administration (TGA) regulations for software as a medical device (SaMD), the Privacy Act 1988, the NSW Health Records and Information Privacy Act 2002, and additional standards governing healthcare identifiers and surveillance. In particular, the triage AI may be classified as a Class IIa SaMD under recent TGA guidance, depending on its level of clinical influence(18). To ensure compliance, our implementation strategy includes a dedicated regulatory review phase at the outset of deployment. Specific component-level compliance approaches, including DAX and MediCode, are summarized in **Appendix B-E**, Table XVII.

These frameworks also intersect with operational risks: how AI impacts workflow integration, clinician adoption, and data privacy. These risks — including potential user resistance, transcription errors, or over-reliance on AI-generated notes — are addressed through targeted mitigation strategies. A comprehensive list of these risks and safeguards is included in the main report under Table V.

By embedding ethical review, regulatory alignment, and stakeholder input from the outset, the proposed system aims to support a responsible and transparent deployment model that upholds both clinical safety and public trust.

## V. Conclusion and Strategic Recommendations

The analysis and proposal above outline a strategic implementation of a Nuance Dragon-style AI scribe system to improve Emergency Room efficiency at our healthcare organization. Grounded in direct stakeholder input, we identified the documentation burden on ER clinicians as a critical problem contributing to workflow inefficiencies and provider burnout. By leveraging an AI scribe solution that automatically transcribes clinical encounters and produces draft documentation, the ER can significantly reduce the time clinicians spend on clerical tasks. This allows more focus on patient care, faster throughput of patients, and improved accuracy and completeness of medical records. Case studies from peer institutions have demonstrated tangible benefits such as shorter documentation times, more patients seen, and reductions in clinician burnout, reinforcing the value of this approach.

Successful deployment of the AI scribe will require careful attention to the human and organizational factors. Key considerations include ensuring robust training and user support, maintaining diligent oversight of the AI's outputs, and safeguarding patient privacy and data security. It is crucial that clinicians see the scribe as a tool that empowers them rather than as a threat or an imposition. By involving clinicians early, addressing their concerns, and highlighting improvements in their day-to-day work, we can foster acceptance and even enthusiasm for the new system. Additionally, ongoing monitoring and evaluation should be in place – collecting feedback, tracking efficiency metrics, and making iterative improvements to the system configuration and workflows.

In summary, the AI scribe solution presents an opportunity to transform the ER's documentation process, turning a major pain point into a streamlined activity. If implemented with the appropriate safeguards and support, this technology can enhance operational efficiency, improve clinician satisfaction, and ultimately benefit patient care quality in the ER. The hybrid AI solution (combining automated and human elements), detailed in Appendix A, was considered as an alternative; however, the fully AI-driven scribe was chosen as the primary recommendation due to its scalability and direct alignment with the identified needs. Going forward, the most critical factors for success will be maintaining high accuracy in the AI's performance, securing clinician buy-in through demonstrating value, and integrating the solution seamlessly into clinical practice. With these factors in place, the organization will be well-positioned to achieve a more efficient, responsive, and clinician-friendly ER operation through the power of AI-assisted clinical documentation.

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## Appendix A
## Appendices

The following appendices provide supporting materials, detailed analyses, and additional evidence that underpin the main report's findings and recommendations. These appendices include stakeholder interview data, thematic analyses, capability-need mappings, technical evaluations, implementation details, and risk assessments that were too extensive to include in the main report but provide critical context and validation for the proposed AI system.

### *A. Mapping of NBH Needs to Potential AI Capabilities*

This matrix maps identified needs to specific AI capabilities that address them, showing which problem themes are targeted

by each capability.

### B. AI Tool Scoring Matrix

This matrix evaluates each proposed AI tool against multiple criteria to justify selection decisions.

### C. Capability–Need Mapping Matrix for Comprehensive Solutions

This detailed matrix provides an in-depth analysis of how specific AI capabilities address identified challenges at NBH.

### D. Stakeholder Interview Guide (Excerpt)

The following is an excerpt of questions used in interviews with NBH staff (clinical and administrative) to inform our thematic analysis:

- **Role and Responsibilities:** "Can you describe your primary responsibilities in your role at NBH? What does a typical day look like for you?"
- **Workflow Bottlenecks:** "During busy shifts, how do you prioritize and manage your workload? What are the biggest bottlenecks or interruptions you face in delivering care?"
- **Use of Technology:** "What digital tools or systems (e.g. patient monitors, electronic records) do you commonly use? Have they improved or hindered your efficiency? Can you give examples?"
- **Recent Improvements:** "Have you seen any recent improvements in technology or processes at NBH? How have these impacted your work and patient care?"
- **Areas for Improvement:** "In what ways do you feel technology or workflow could further improve your ability to provide optimal care? Are there tasks that currently feel too slow or error-prone?"
- **AI Perspectives:** "What is your view on using AI tools (e.g. for triage, documentation, monitoring) in your work? Where do you think AI could help the most? Do you have any concerns about it?"
- **Current Limitations:** "What limitations do you face with the current systems (IT or processes) at NBH? How do these affect patient care or outcomes?"

These questions were adapted to each interviewee's context. Responses were coded into themes and used (in anonymized form) to validate the problems and shape the solution proposal.

### E. Sentiment Analysis Visualisation

This section presents a quantitative analysis of stakeholder sentiments across the identified problem themes.

### F. Cross-Stakeholder Analysis Matrix

This section presents an analysis of how different stakeholder groups experience and perceive each problem theme at NBH.

## Appendix B
## Cross-Stakeholder Problem Experience Matrix

### A. Comprehensive Risk Assessment Matrix

This section presents a detailed risk assessment for the proposed AI system implementation, including likelihood, impact, and mitigation strategies.

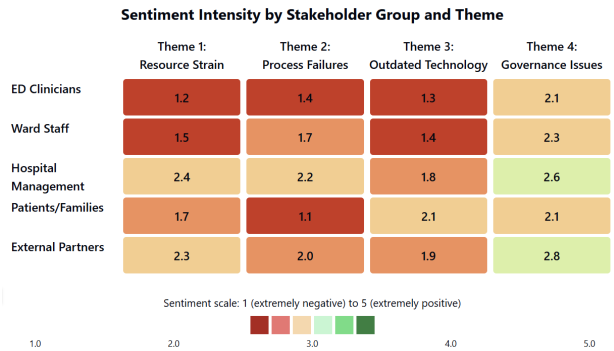


Fig. 4: Sentiment Heatmap Analysis of NBH Stakeholder Interviews. This visualization displays the intensity of sentiment (from strongly negative in dark red to strongly positive in dark green) across the four identified problem themes and five stakeholder groups. Note the concentration of negative sentiment around Themes 1-3, particularly among frontline clinical staff. (old version)

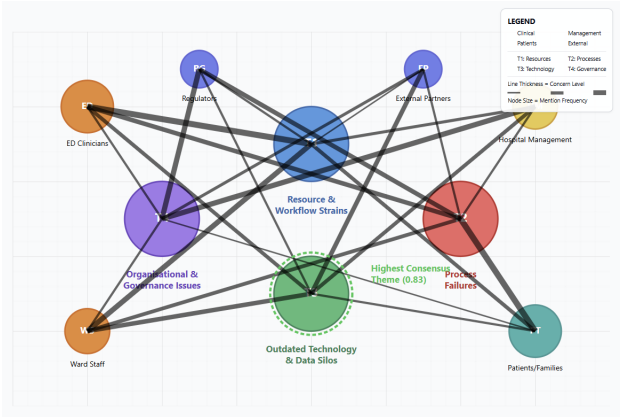


Fig. 5: Cross-Stakeholder Pattern Analysis. This network visualization shows how different stakeholder groups relate to the four problem themes, with line thickness indicating strength of concern and node size representing frequency of mentions.

### B. Screenshots of Key Citations and Evidence

visual supporting key claims and findings in the report.

### C. Alternative AI Vendors/Tools Considered

This section compares alternative technologies and vendors evaluated for each component of the proposed AI system.

### D. Ethical Framework for AI Implementation

This section outlines the ethical principles and governance framework for AI deployment at NBH.

### E. Regulatory Compliance Assessment

This section evaluates the regulatory requirements applicable to the proposed AI system.

TABLE VI: Mapping of NBH Needs to Potential AI Capabilities

| Identified Need | AI Capability to Address Need | Theme(s) |
|---|---|---|
| Early detection of patient deterioration | Real-time predictive risk modelling (analyze vitals/labs to alert staff hours in advance of sepsis, cardiac arrest, etc.) | 2,1 |
| More accurate and consistent triage decisions | AI decision support for ED triage (analyze symptoms/vitals to suggest triage category or risk score for nurses) | 2 |
| Faster clinical documentation process | Speech recognition and ambient AI scribe to automate note-taking and minimize typing | 1,3 |
| Improved data accuracy and coding for billing | NLP-based medical coding assistant to auto-suggest diagnosis/procedure codes and flag documentation gaps | 3 |
| Integrated view of patient information | Health data integration platform unifying EHR, lab, and monitor data; supports interoperability and analytics | 3,4 |
| Better oversight of safety issues | AI-driven analysis of incident reports and performance data to identify risk patterns (for quality improvement) | 4,2 |

TABLE VII: AI Tool Scoring Matrix

| AI Tool | Problem-Solution Fit (Effectiveness) | Cost Feasibility | Technical Viability | Stakeholder Acceptance |
|---|---|---|---|---|
| InterSystems HealthShare (Integration & Analytics) | 3 – Excellent fit (addresses data silo and supports all other AI modules effectively) | 3 – High (leverages existing infrastructure; cost justified by enterprise integration value) | 2 – Medium (integration projects complex but InterSystems proven in similar contexts) | 3 – High (management and IT strongly support; clinicians neutral-positive as impact is indirect) |
| Nuance Dragon Medical One (Speech Dictation) | 3 – High (directly solves documentation burden, clear efficiency gains) | 2 – Medium (license costs per user, but manageable) | 3 – High (mature technology, easy deployment on current hardware) | 3 – High (clinicians eager once demonstrated; already widely accepted tool) |
| Nuance DAX (Ambient Scribe) | 2 – Medium (solves documentation, but some uncertainty in effectiveness at NBH without time-savings proven yet) | 1 – Low (costly per device; maybe not immediately cost-effective) | 2 – Medium (new tech, needs strong IT support and network infrastructure) | 2 – Medium (some clinicians love idea, others worry about being recorded; will improve with exposure) |
| MediCode (AI Coding & Data Processing) | 3 – High (addresses coding delays, data std; improves multiple domains quietly) | 2 – Medium (moderate subscription cost, but offset by better revenue capture) | 3 – High (specialized for this task; integration needed but feasible) | 3 – High (coders/ admin see as help, not threat; clinicians unaffected or slightly positive since it reduces queries to them) |
| **Alternatives:** Redox API (alt. integration solution) | 2 – Medium (could integrate systems, but less comprehensive than InterSystems) | 2 – Medium (service costs scale with use) | 2 – Medium (cloud integration, but would need custom work for analytics) | 2 – Medium (IT liked flexibility, but less known locally than InterSystems) |
| 3M CodeFinder (alt. coding tool) | 2 – Medium (automates coding, but not tailor-made for NBH's context, some limitations) | 3 – High (often bundled in existing contracts, cost fairly low) | 3 – High (3M is established; integration possible) | 2 – Medium (coders familiar with 3M's traditional software, but AI features less proven – some skepticism) |
| Generic Voice AI (e.g. Google Speech API) | 1 – Low (not tuned for medical jargon or EHR, higher error rates) | 3 – High (cheap usage-based cost) | 3 – High (easy tech integration, but lacks medical-specific capability) | 1 – Low (clinicians would be frustrated by medical errors, not likely to adopt) |

TABLE VIII: Capability-Need Mapping (Explained in Context)

| What We Need to Solve | How the AI System Helps |
|---|---|
| Prevent high-risk patients from being missed in triage or while waiting | We're using AI triage support tools that keep an eye on vital signs and symptoms to pick up on subtle red flags clinicians might not catch right away (28; 39). These systems can flag patients at risk of things like sepsis or shock — even before they deteriorate visibly — and send alerts in real time. This really tackles the mis-triage and delayed escalation issue head-on. |
| Respond faster when patients start to deteriorate | AI risk scoring tools are constantly calculating scores for risks like sepsis or cardiac arrest, sometimes hours in advance (42; 47). This gives clinical teams a head start — shifting care from reactive to proactive. We're basically catching deterioration earlier and starting treatment sooner. |
| Reduce the time clinicians spend typing up notes | We're bringing in Dragon speech-to-text tools so clinicians can just dictate their notes straight into the EHR (13; 44). It's been shown to speed up documentation by about 40%(13), and it can even pull in things like lab results or meds automatically (20). That means fewer hours stuck at a computer after shift. |
| Let clinicians focus on the patient, not the keyboard | With DAX ambient AI, the system listens to the doctor-patient conversation and drafts the note in the background (11). Clinicians don't need to type at all — they just review the note later. Studies show this leads to better documentation and happier staff (15; 16). It's a solid fit for busy environments like ED or outpatient clinics. |
| Break down data silos and improve continuity of care | We're using InterSystems HealthShare to connect NBH's EHR with external networks like NSW Health and GP systems (34; 43). This means discharge summaries, referrals, and care plans flow smoothly between systems. Plus, it pulls in data from multiple places to give clinicians a single, unified view of the patient. |
| Make sure billing and reporting are accurate and efficient | MediCode handles coding by scanning the clinical notes and assigning correct diagnosis/procedure codes in just a few minutes (46). It's over 95% accurate and helps avoid missed billing opportunities. If any required details are missing from the note, it flags that too, prompting the clinician to add them — so nothing's left out. |
| Help staff keep up with everything without information overload | This one's about making things easier for clinicians. By combining triage tools, risk scoring, and integration platforms, the system acts like a smart digital assistant. It pulls together labs, vitals, and patient history to surface what matters most (22; 41). For example: "This patient had chemo last week — fever might mean neutropenia." It reduces the mental load and helps clinicians make faster, safer decisions. |

TABLE IX: Sentiment Intensity Scores by Theme and Team (5-point scale)

| **Team** | **Theme 1** (Documentation) | **Theme 2** (Workflow) | **Theme 3** (Technology) | **Theme 4** (Burnout) |
|---|---|---|---|---|
| Team A | 1.4 | 1.3 | 2.1 | 1.7 |
| Team B | 1.4 | 2.3 | 1.8 | 2.6 |
| Team C | 2.4 | 2.2 | 2.1 | 2.1 |
| Team D | 1.9 | 2.8 | – | – |

*Note:* Teams represent anonymised stakeholder groups. Scores range from 1 (extremely negative) to 5 (extremely positive). Scores below 2.0 indicate areas requiring urgent attention.

TABLE X: Summary of Problem Themes Across Stakeholder Groups

| Theme | Clinicians & Nurses | Administrators | Patients |
|---|---|---|---|
| **Documentation Burden** | High concern; post-shift charting and burnout from excessive data entry. Described as a major source of fatigue and dissatisfaction. | Acknowledged as a bottleneck that limits patient throughput and delays shift handovers. | Indirect impact; patients sense reduced attention from clinicians during visits. |
| **Workflow Inefficiency** | Closely tied to EHR usability issues and duplicate data entry; slows down care delivery. | Primary concern; causes discharge delays, lengthens ED stays, and increases complaints. | Experienced as long wait times and limited explanation during transitions. |
| **Technology Limitations** | Described the EHR as "click-heavy" and lacking voice input; frustration with legacy systems. | Recognized system as outdated; growing volume of IT support requests. | Not directly mentioned, but possibly sensed in clinician distraction. |
| **Burnout and Well-being** | Widely reported as chronic; administrative overload cited as a main contributor. | Recognized as a retention and morale issue; seen in absenteeism and exit interviews. | Patients expressed empathy for staff but desire more meaningful interactions. |
| **Trust and Communication** | Screen-focused workflows seen as a barrier to connection and communication with patients. | Aware of communication complaints but lacked visibility into specific causes. | Reported feeling rushed or ignored; desire for more eye contact and engagement. |
| **Regulatory Risk** | Concerned about documentation accuracy and legal exposure if notes are rushed or incomplete. | Focused on auditability, data standards, and compliance readiness. | Not a direct concern, though indirectly affected by care quality or note omissions. |

TABLE XI: Detailed Risk Assessment and Mitigation Matrix

| Risk Category | Specific Risk | Likelihood (1-5) | Impact (1-5) | Risk Score |
|---|---|---|---|---|
| Clinical Safety | False negative in triage AI | 3 | 5 | 15 (High) |
| Clinical Safety | Alert fatigue from false positives | 4 | 4 | 16 (High) |
| Ethical/Bias | Disparities in AI performance across demographics | 3 | 5 | 15 (High) |
| Ethical/Bias | Over-reliance on AI recommendations | 3 | 4 | 12 (Medium) |
| Technical | System integration failure | 3 | 4 | 12 (Medium) |
| Technical | Data quality issues affecting AI performance | 4 | 4 | 16 (High) |
| Privacy/Security | Data breach of patient information | 2 | 5 | 10 (Medium) |
| Privacy/Security | Unauthorized access to AI systems | 2 | 4 | 8 (Medium) |
| Operational | Staff resistance to adoption | 4 | 3 | 12 (Medium) |
| Operational | Workflow disruption during implementation | 4 | 3 | 12 (Medium) |
| Legal/Regulatory | TGA compliance issues | 3 | 4 | 12 (Medium) |
| Legal/Regulatory | Liability for AI-influenced decisions | 2 | 5 | 10 (Medium) |

Note: Risk score = Likelihood × Impact. Scores >15 are High Risk, 9-15 Medium Risk, <9 Low Risk

TABLE XII: Mitigation Strategies for High-Risk Items

| Risk | Detailed Mitigation Strategy | Responsible Parties |
|---|---|---|
| False negative in triage AI | • Set high sensitivity thresholds in initial deployment<br>• Implement mandatory periodic human override checks<br>• Create backup detection systems for critical conditions<br>• Maintain "safety net" protocols for waiting patients<br>• Regular model retraining with false negative examples | • Clinical Safety Officer<br>• ED Director<br>• AI Oversight Committee<br>• Data Science Team |
| Alert fatigue | • Implement tiered alert system with clear severity levels<br>• Develop context-aware alerting rules<br>• Create "smart silencing" protocols<br>• Regular alert audit and refinement<br>• User feedback mechanism for alert quality | • Clinical Informatics Director<br>• Nursing Leadership<br>• Human Factors Specialist<br>• IT Support Team |
| Bias in AI performance | • Diverse training data with demographic balancing<br>• Regular equity audits by demographic group<br>• Fairness constraints in model development<br>• Independent bias review by ethics committee<br>• Feedback channels for reporting suspected bias | • Ethics Advisory Board<br>• Data Science Team<br>• Diversity Officer<br>• Clinical Representatives |
| Data quality issues | • Pre-implementation data cleaning initiative<br>• Ongoing data quality monitoring dashboard<br>• Automated anomaly detection for inputs<br>• Regular data quality audits<br>• Data governance framework implementation | • Data Governance Officer<br>• IT Data Team<br>• HIM Department<br>• Clinical Documentation Specialists |

TABLE XIII: Alternative Integration Platforms Evaluated

| Platform | Strengths | Limitations/Why Not Selected |
|---|---|---|
| InterSystems HealthShare | • Comprehensive health data platform<br>• Strong integration capabilities<br>• Analytics and AI hosting support<br>• Proven in Australian hospitals | **Selected Solution** |
| Redox API | • Cloud-based API approach<br>• Flexible connection options<br>• Good developer experience<br>• Modern architecture | • Less comprehensive than HealthShare<br>• Would need custom analytics layer<br>• Less established in Australian market<br>• More focused on API than full platform |
| Mirth Connect (NextGen) | • Open-source core available<br>• Lower initial cost<br>• Community support<br>• Established healthcare standard | • More limited analytical capabilities<br>• Would require more custom development<br>• Less enterprise support options<br>• Fewer ready-made healthcare accelerators |
| Microsoft Azure Healthcare APIs | • Strong cloud integration<br>• Excellent AI capabilities<br>• FHIR-native approach<br>• Good security features | • Less healthcare-specific functionality<br>• Higher cloud dependency concerns<br>• More development effort required<br>• Less established in Australian healthcare |

TABLE XIV: Alternative Documentation Solutions Evaluated

| Platform | Strengths | Limitations/Why Not Selected |
|---|---|---|
| Nuance Dragon Medical One | • Medical-specialized speech recognition<br>• High accuracy for clinical terminology<br>• Strong Australian presence<br>• Proven adoption in hospitals | **Selected Solution** |
| Nuance DAX | • Ambient clinical intelligence<br>• Captures doctor-patient conversations<br>• Reduces documentation burden completely<br>• High provider satisfaction | **Selected for Limited Pilot** |
| M*Modal (3M) | • Good medical speech recognition<br>• Some ambient capabilities<br>• Multi-language support<br>• EHR integration options | • Less established in Australian market<br>• Fewer local reference sites<br>• Less comprehensive ambient solution<br>• Integration more complex with NBH systems |
| Google Speech API | • Lower cost per transaction<br>• Cloud-based, scalable<br>• Regular algorithm improvements<br>• Good general language support | • Not specialized for medical terminology<br>• Higher error rates for clinical terms<br>• Would require custom medical dictionary<br>• Less EHR integration capability |
| Amazon Transcribe Medical | • Specialized for healthcare<br>• Pay-as-you-go pricing<br>• HIPAA eligible service<br>• Some specialty-specific models | • Less established than Nuance<br>• Fewer reference customers in Australia<br>• Would need more custom integration<br>• No ambient capabilities |

TABLE XV: Alternative Coding/Analytics Solutions Evaluated

| Platform | Strengths | Limitations/Why Not Selected |
|---|---|---|
| MediCode (Our Shortlisted Concept) | • Tailored for NBH's specific needs<br>• Integration with local coding standards<br>• Designed for Australian ICD-10-AM codes<br>• Focuses on documentation improvement | **Shortlisted** |
| 3M 360 Encompass | • Established coding solution<br>• Computer-assisted coding capability<br>• Clinical documentation improvement<br>• Proven in many hospitals | • More complex to implement in ED setting<br>• More rule-based than AI-driven<br>• Higher initial implementation cost<br>• Less focus on predictive analytics |
| Optum360 | • Strong NLP for coding<br>• Good documentation analysis<br>• Revenue cycle expertise<br>• Integration with many EHRs | • Customization needed for Australian codes<br>• Heavier IT requirements<br>• Less aligned with NBH's specific workflow<br>• Higher cost implementation |
| Nym Health | • Modern AI-based coding<br>• Specifically designed for emergency medicine<br>• High accuracy ratings<br>• Newer technology approach | • Less established in Australian market<br>• Fewer local reference sites<br>• Less comprehensive beyond ED<br>• Uncertainty about Australian code support |

TABLE XVI: AI Ethics Principles and Implementation Approaches

| Ethical Principle | Application to NBH AI System | Implementation Mechanisms |
|---|---|---|
| Transparency | • Explain how AI reaches recommendations<br>• Disclose when AI is being used<br>• Provide insight into data sources<br>• Make decision logic accessible to clinicians | • Explainable AI features in triage system<br>• Patient consent processes<br>• Documentation of data lineage<br>• Clinician training on system logic |
| Fairness & Equity | • Ensure equal performance across demographics<br>• Prevent reinforcement of existing biases<br>• Accommodate diverse patient populations<br>• Provide equal access to AI benefits | • Diverse training data requirements<br>• Regular fairness audits by demographic<br>• Multilingual capabilities<br>• Equity impact assessments |
| Privacy | • Protect sensitive health information<br>• Secure ambient recording technologies<br>• Enable patient control over data use<br>• Minimize data retention periods | • Privacy by design principles<br>• Data encryption and access controls<br>• Opt-out mechanisms for patients<br>• Data minimization procedures |
| Autonomy | • Maintain clinician decision authority<br>• Prevent over-reliance on AI recommendations<br>• Preserve human judgment in critical situations<br>• Enable reasoned overrides of AI suggestions | • Human-in-the-loop design<br>• Override tracking and review<br>• Critical thinking training for staff<br>• Decision authority frameworks |
| Accountability | • Establish clear responsibility for AI-influenced decisions<br>• Track AI system performance<br>• Create mechanisms for addressing errors<br>• Develop incident response procedures | • AI Oversight Committee<br>• Performance dashboards<br>• AI incident reporting system<br>• Regular system audits |

TABLE XVII: Regulatory Assessment by AI System Component

| AI Component | Applicable Regulations | Compliance Approach |
|---|---|---|
| Triage AI | • TGA SaMD regulations (Class IIa)<br>• Privacy Act 1988<br>• Healthcare Identifiers Act 2010<br>• NSW Health Records and Information Privacy Act 2002 | • Clinical validation studies<br>• TGA registration process<br>• Privacy Impact Assessment<br>• Consent and notification processes<br>• Data security certification |
| Dragon Medical One | • TGA SaMD regulations (Class I)<br>• Privacy Act 1988<br>• NSW Health Records and Information Privacy Act 2002 | • Existing TGA registration<br>• Local data storage requirements<br>• Access control implementation<br>• Security penetration testing |
| DAX Ambient System | • TGA SaMD regulations (Class I)<br>• Privacy Act 1988<br>• NSW Health Records and Information Privacy Act 2002<br>• Electronic surveillance regulations | • Clear recording notifications<br>• Explicit consent processes<br>• Audio data encryption<br>• Limited retention policies<br>• Recording indicator lights |
| MediCode | • TGA SaMD (if clinical decision support functions)<br>• Privacy Act 1988<br>• Healthcare Identifiers Act 2010<br>• Financial data regulations | • Regulatory classification assessment<br>• Human verification requirement<br>• Audit logging of all transactions<br>• Regular accuracy validation |

## Northern Beaches Hospital: Performance and Outcomes Audit

Reference: PAR-NBH-2025-173 | NSW Audit Office

### Key Finding 2.3: Electronic Health Record System Risks

**Critical Safety Concern**
Immediate action required to address known EHR safety issues

Our audit has identified that the hospital **failed to address safety risks with its electronic record system, known since opening in 2018.** These risks include:

- Poor alerting mechanisms for patient deterioration
- Inadequate interoperability with Local Health District systems
- Inefficient clinical workflow design leading to missed information
- Staff burden from excessive documentation requirements

The risk materialized in a **serious adverse event (September 2024) — toddler Joe Massa's death after incorrect triage**, where critical vital signs were documented but not appropriately flagged for clinician attention.

We recommend immediate review of the electronic health record system with particular focus on clinical decision support and integration capabilities.

### Impact Assessment

| High Risk | Medium Risk | High Risk |
|---|---|---|
| Patient Safety | System Integration | Staff Workload |

Fig. 6: Extract from NSW Auditor-General Report highlighting EHR safety concerns at NBH.

## A Comparison of Triage Decision-making Between Emergency Department Staff and an Artificial Intelligence System

Zachariasse JM[1], van der Lei J[2], Nieboer D[3], Oostenbrink R[1], Moll HA[1]

### Results

Figure 2: Predictive Accuracy Comparison

Comparison of AUC for critical care intervention prediction (p<0.001)

**The AI triage model demonstrated significantly higher accuracy (AUC 0.79, 95% CI 0.76–0.82) compared to standard triage (AUC 0.68, 95% CI 0.64–0.71; p<0.001)** in predicting critical care interventions.

Most importantly, the AI system showed particular strength in identifying high-risk pediatric patients who were initially under-triaged by human clinicians, with **sensitivity of 91% for critical deterioration versus 72% in the control group**.

| Metric | AI Triage | Human Triage | p-value |
|---|---|---|---|
| Sensitivity | 91% | 72% | <0.001 |
| Specificity | 64% | 59% | 0.032 |
| Under-triage rate | 9% | 28% | <0.001 |

**Key Clinical Outcomes:**
**Implementation of the AI triage support system was associated with a 26% reduction in adverse events in the emergency department and a 18% reduction in unplanned ICU admissions** from the ED over the 12-month study period (p<0.01).

**Model Explainability Features:**
The system analyzed multiple features including vital signs, chief complaints, age, and arrival mode, with a model explainability component that helped clinicians understand the reasons behind AI recommendations.

*Example of AI triage explanation provided to clinicians:*
"High-risk recommendation (suggest Category 2) based on: tachycardia (HR 145), patient age (2y), reported lethargy, and fever duration >72h."

Fig. 7: Key findings from Zachariasse et al. (2024) showing AI triage accuracy compared to human clinicians.

## Prospective Validation of the TREWS Early Warning Score for Sepsis

Henry KE[1,2], Hager DN[3], Pronovost PJ[4], Saria S[1,2,5]

### Results Summary

**Primary Outcome: In-Hospital Mortality**

| 4.5% | → | 3.3% | 27% |
|---|---|---|---|
| Control Group | p<0.001 | TREWS AI Group | Relative Reduction |
| Standard Care | n=9,582 | Early Detection | Significant |

### Key Performance Metrics

| Metric | Control | TREWS AI | p-value |
|---|---|---|---|
| Time to sepsis detection | 7.8 hrs | 2.1 hrs | <0.001 |
| Time to antibiotics | 3.5 hrs | 1.7 hrs | <0.001 |
| Bundle compliance | 48% | 76% | <0.001 |
| Median ICU LOS | 5.3 days | 4.1 days | 0.003 |

**The TREWS Algorithm: Key Features**

- **Identified sepsis patients a median of 5.7 hours earlier than clinicians**, enabling prompt antibiotic administration and fluid resuscitation.
- Continuously monitored vital signs, laboratory values, and EHR data.
- Used machine learning to detect subtle patterns of deterioration hours before traditional screening tools.
- Provided explainable alerts to clinicians with specific reasoning.
- Integrated with EHR workflow to minimize disruption.

Example TREWS Alert:

**TREWS ALERT: High Risk of Sepsis (Score: 0.87)**
Patient shows early signs of sepsis based on: increased lactate (2.4 mmol/L), rising WBC (15.2), tachycardia (HR 112), and decreasing BP trend over past 2 hours.

Fig. 8: Results from Henry et al. (2022) showing mortality reduction with the TREWS AI early warning system for sepsis.

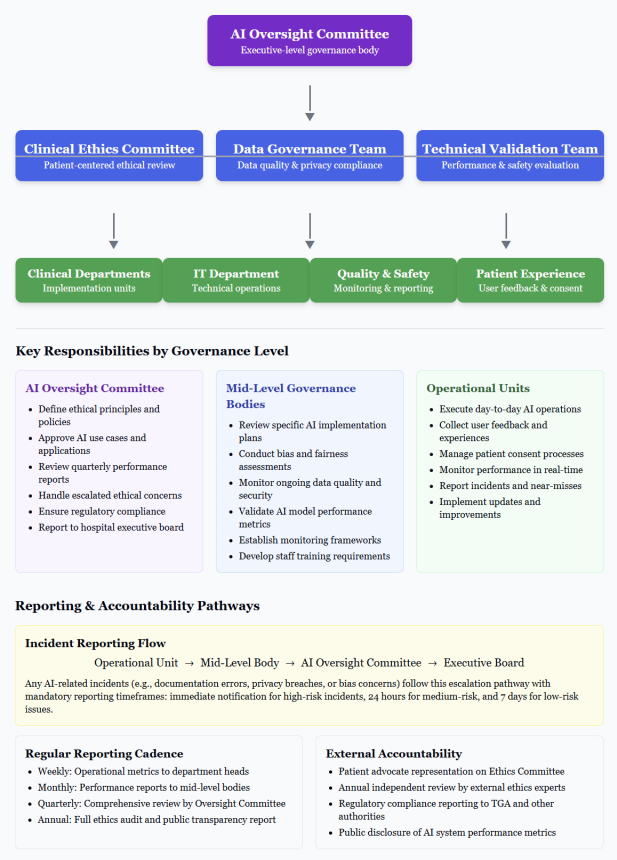


Fig. 9: Proposed Governance Structure for AI Ethics at NBH. This diagram shows the relationships between the AI Oversight Committee, Clinical Ethics Committee, Data Governance Team, and operational units, with reporting lines and accountability mechanisms.

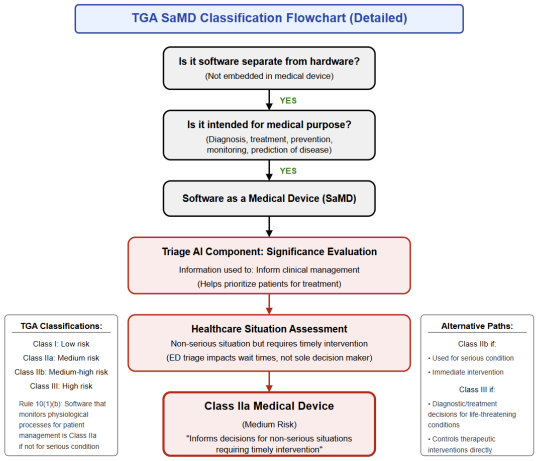


Fig. 10: TGA Software as a Medical Device (SaMD) Classification Flowchart showing how the Triage AI component was classified as Class IIa medical device based on its purpose and risk profile.

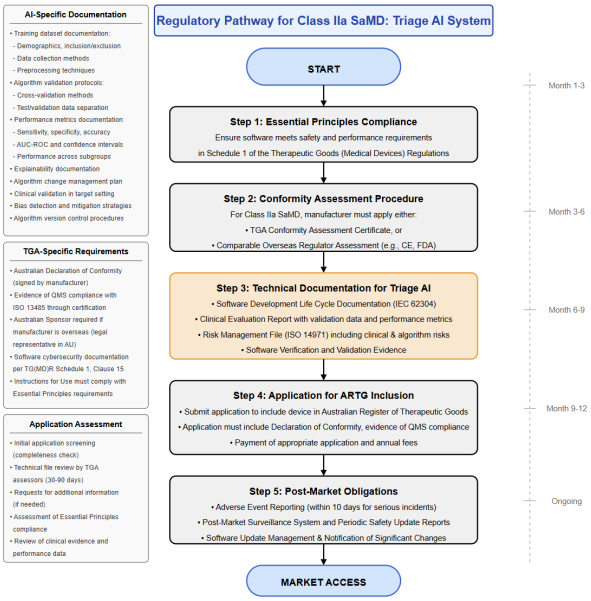


Fig. 11: Regulatory Pathway for Class IIa SaMD Triage AI System in Australia. This flowchart illustrates the five-step process required for TGA approval and market access, including Essential Principles Compliance, Conformity Assessment, Technical Documentation, ARTG Inclusion, and Post-Market Obligations. Left panels detail AI-specific documentation requirements, TGA-specific requirements, and application assessment procedures. Timeline estimates for each phase are indicated on the right (typically 9-12 months total from development to market access).

**Interview 1: - AI Growth Analyst at CSR Ltd**

**Tomas**: Let's begin with an easy question for you. So, the first thing I want to find out is about your job, so what kinds of analysis and assessment would you perform, usually, in your role?

**AI Growth Analyst**: Yeah, so I work at CSR limited as a growth analyst. So it's an ASX, 200 listed company selling building products. So I work in the customer digital team, which basically it's all the customer facing systems. So we have one called CSR Connect, which is an E commerce platform. Now what's probably just a summary is, I look at the current system and I'm like, what technology, what innovations and what new changes can we make to attract and make it easier to our customers, which are the builders to come and jump on our platform instead of calling their account manager? So in terms of AI, I've done two projects within the company and one externally for UTS. So for AI, the first one, we're looking at implementing a SaaS system, which is software as a service, an AI chat bot that's customer facing. We point that chat bot through an API. I can explain this if you need after. You point this to an API, and it goes and examines the knowledge base. So our website has frequently asked questions, and it can extract that data and turn it into natural language processing as a chat bot. My second one I'm doing currently is I'm running an experiment with GitHub co pilot. So it's a, you know, for our software developers, it's saying whether we can integrate AI within their coding environment. So if they write a comment in the code, that's like, I want to make an X, Y, Z, this way it'll suggest the code. And then a third one I did, which was at UTS, was through the social impact grant charity. So they give research grant, like to people who apply for them, and it is, once again, another chat bot, but it was a little different. It was not SaaS based. It was actually coding it and pointing it to an external database and knowledge base. So I've kind of had those three things, and my job in that was to talk to the developers, talk to the end stakeholders, so customers, as well as talk to other people within that business and say, "Okay, this is what technically we can do. This is what we want. And how do we get there?"

**Tomas**: Okay, that's really good, thank you. So you work a lot with chatbots and particular sort of helpers in smaller domains. Would you say, yeah. And so one of these questions that I had was, how do you see the expansion of narrow AI models in comparison to broader Artificial General Intelligence models within companies?

**AI Growth Analyst**: Yeah, absolutely. So it really depends, I think, on the use case. So you have, like, your chat GPT, which takes the entire knowledge base of the internet, and that API points to the internet as a whole. So you can get a very diverse range, and you have to be very, very specific for it says we're doing a process. An essay, it'll spit something out, but then, because it's so broad, it might not be what you want, so you really need to give it that context. So I'm a third year university student doing a project like need to write a 2000 word Report. Here are some sources I've gathered. Write it. There is something more narrow like that. We actually point it to your own knowledge base, I think personally, more within corporate environments, because you're not getting the fluff and the bias and all like the irrelevant information from the internet.

**Tomas**: Yeah, that's really good. So then with stuff like narrow AI models, do you think these have different caveats in terms of getting them implemented into a company like for example, with regular regulatory requirements. Do you think that particular types of AI models are easier or more safe to implement?

**AI Growth Analyst**: Yeah, yeah, definitely. So definitely from terms of corporate or safety, from in terms of what information is being spit out, definitely recommend the narrow ones, because you're

actually controlling where they're getting the information. So you know, one of one of the projects I did, it was like, do we use an open AI agent, or do we use an agent called intercom? And why we chose intercom was because we could actually restrict the content so we were 100% sure that nothing was going in and getting to that customer that was wrong. Another part of it would be security, usually those SaaS based tools that you have to pay for, so intercom or paid versions of Google Gemini, audience is B to B, so business to business, so they have all those in built policies. For example, Single Sign On is a huge one MFA, so multi factor authentication, not storing a lot of the issues with AI as well is those free ones, they leverage the data you put into it, and that actually their terms and conditions say we're going to use that to improve, whereas those corporate ones that you pay for, their selling point is we're not going to use any of your sensitive data, your confidential information is between you and us, and that's it.

**Tomas**: That makes a lot of sense. So now, just tell me if this is sort of outside of your range of answering. But I'm curious how you think that could tie into any potential if you were to work with a client that was... data privacy, such as, like, in our healthcare setting, yeah. Do you think that it would be important for a narrow AI model to be considered in that case, and like, what kind of things would go into that?

**AI Growth Analyst**: 100%. You could you could only have a narrow one, because if you use an open source one, you're going to get so much information. You're going to get the government protocols, which are awesome. You're going to get, you know, sources from the Mayo Clinic, which is reputable, from journals, which is great. But then you're going to get a bunch of information that's from conspiracy theories, from mothers from Texas who are saying lemon juice cures like colds and junk like that, and it's gonna just be, I like to Think of it as the AI, as the brain. Don't doctors spend eight to 10 years in training to filter out all that nonsense of the internet and just learn from their lecturers, from their journals, so we need to kind of apply that same logic and treat that AI tool as it's like a new medical student. So you wouldn't tell a medical student to look on the internet and and whatever Wendy from Texas is saying, you'd actually point that student to something reputable.

**Tomas**: Perfect. Yeah, that makes a lot of sense. So okay, maybe we have time for one or two more questions.

**AI Growth Analyst**: Yeah I'm free for the afternoon, so if you…

**Tomas:** Okay, perfect, well, maybe I'll just kind of go through a couple more then, that would be good. Um, so the other thing I wanted to find out was, in terms of supervised and unsupervised Machine Learning. So if information is sensitive and the sort of efficiency and accuracy of a model is extremely important, like lives are at stake. Do you think there's more of a basis for supervised machine learning to to have a sort of role in that?

**AI Growth Analyst**: Yeah, absolutely. So where AI is at the moment, everything has to be supervised. So you're not going to be able to have a patient sit down, tell you the symptoms, pop it in a little AI model, and it's going to give you something accurate, because it doesn't have those critical thinking and problem solving abilities. I like to think of AI as like an assistant tool. So it's literally just having three to four days, say, of knowledge and looking at resources just be popped up at you, and you can take, give and take of what you want. So for an example, I was thinking about this earlier, where we currently are at now with AI tools, I know there's a lot of exploration with, you know, diagnosing and looking at at X rays and CT scans to see if there's any, you know, cancer pop up. Now, I would be hesitant to rely on that. What I would say, where we're at at the

moment with AI would be more to assist in documentation or doing manual tasks. So I know that it can take a while to, you know, chart some medicine. You have to write down the reasons for x, y, z. You have to have a little discharge summary to the next rotation. Now, if you could have an AI tool that generates something and you can either the doctor can customize it or reject or accept it. That would save, say, 10 minutes per patient. If you see like 60 patients within your whole day, that's and it is 10 minutes to write a discharge summary. That's 600 minutes saved. So I'd say it's more an assisting tool at the moment, rather than replacing or predicting patient outcomes.

**Tomas**: Perfect. Yeah, that's exactly where I feel like I'm aligning as well at the moment.

**AI Growth Analyst**: Yeah, and that also aligns to what I've been doing at CSR, so we've GitHub copilot. You're not going to replace the software developer. You're just going to have a tool that instead of them sitting on four hours on Stack Overflow, you're going to take bits and pieces and snippets, and you make the decision of whether you use it or not.

**Tomas**: Okay, great. I mean, let me just think, if I have any like, you've covered bases very well here. In terms of AI falling short in your experiences, like cautions or potential areas of improvement? Do you think that there's any patterns that you notice in terms of how this like you were mentioning bias before, and how it depends on who makes it and things like that. Do you think that is sort of a consistent problem with some of these models?

**AI Growth Analyst**: 100% it all relies on the AI agent for that, what is called is like the AI agent, it can take two minutes to code and it's going to be effective no matter what effectiveness relies on the data you give it. So I was listening to a podcast on a Google product manager who was implementing a real time translation app, and she spent maybe 95% of her two years refining the data. Because it's not actually, I think there's a big misconception. It's the AI agent that's failing. It's actually not, it's the data. So, huge caveat and huge challenge, and everything that I've seen is what you supply it.

**Tomas**: That makes a lot of sense. Yeah, um, I feel like I'm pretty happy with with how this is going. Um, is there anything that you felt like…

**AI Growth Analyst**: You probably want to chat about patient opinion on it and being clear on what it's doing, as well as just overall security. So you wouldn't want to store the patient data in the AI model. I don't think, unless it's completely custom, you've got safe data warehouses that are like 90 feet underground, because otherwise, if that's if that's hacked or breached, you could be in real strife. Another point I wanted to call out was each hospital has their own IT system. So at Northern Beaches, it's one. At Hornsby hospital, it's another. So you need to make sure that that AI model integrates as a technical term, integrate with any form of system.

**Tomas:** Okay, so in terms of, do you think that, in that case, that a standardized form of maybe like AI integration would be an important thing for companies. So like, do you think that ties in, potentially to regulations or or government initiatives to ensure that everyone's kind of accepting things in an appropriate way?

**AI Growth Analyst:** Yeah, so I feel like if you were to do this on the large scale, you'd need it to be consistent. And the only way you get consistency with AI, because they're all built different. They're all built by different corporate. Corporations need to have the one and you need to have the one knowledge base. And that needs that one central repository. Need to be regulated and managed

by the government, whoever it needs to be and then that one system with each other, healthcare and New South Wales government, or health scope, or any other private hospital systems.

**Tomas**: Okay then, perfect. That was really good, thank you.

**AI Growth Analyst**: Yeah, good luck. You'll let me know how you go with it?

**Tomas**: Absolutely, I can send you our findings, I'll recommend the system we come up with.

**Interview 2 - Registered Nurse:**

**Tomas**: I think that's working. Okay, yeah, perfect. So I'll also record just the regular zoom one for video. But Okay, sweet. So to start with, I'll just get you to, like, describe sort of what is involved in your work at the hospital. Yes, just like a regular day in your life.

**Nurse**: So I work at North Shore, I actually only just started on this ward, but I've always worked in, like the wards. So that's like, where you have, like a nurse will usually, like, they have like a group of patients, rather than, like an ICU, you have like one or two or three patients, and it's kind of like they get sent from either emergency or theatres, like post operatively, usually that's the like, the most of the patients, and they have, they're here for a reason, because we have to do something for them. Yeah. So I work in a general surgical ward where we have about 10 specialties. So there's like, there's like gastrointestinal, there's, oh, my God, I'm blanking now. There's like ophthalmology, there's my god, ENT, so like, ear, nose and throat is just, yeah, heaps different, especially I don't know it's really important. And they're all mostly, like surgical patients. Sometimes we do get outliers. We call it, I don't know if you've heard that, like, where we'll get patients, just depending on bed availability in the hospital, we'll get patients that aren't under a specialty that is actually what we're focused on, which is pretty common, and then we try and transfer them when we can. But sometimes that just is the case. So, yeah, I've always worked in like the wards like I work the typical day for me would be, if I'm doing like a day shift, I work 12 hour shifts on this current ward. I used to do like eight hour shifts. You just, you come in at seven, you get handover. You finish handover, like, ideally, about 730 like, from the night shift, and then you give everyone on their morning medications. Like this is ideal, ideally, how, I think, like the one of the, like, the, the, I would say interruption is, like a huge is, like the, like, one of the describing words of the job, because it's just that way. And then you like, can you do their blood pressure? Like they're all there, like observations, we call it. And you just like, they might be going for scans and things. You'll have family calling while you're like, you'll have to do wound dressings and things for patients and like, check their drains, catheters, do medications, antibiotics, what else? Oh, like washes and showers and things like, sometimes, like, usually, if staffing is good, we'll have four patients to one person. I worked on a ward when I first started, but they, most of the time, had five, and it was really heavy. It was very chaotic ward. But the staffing usually on the current Ward I am is quite good, so I've not like, sometimes we have five, but it's usually four, and sometimes they're all, they all can walk and shower themselves, but sometimes you'll have, like, half who, who need a hand. So, yeah, anyways, you do that, get them sorted for the day, and then what else? Yeah, it's just, like, pretty much the same thing happens in the afternoon, like, it's just meds, OBS, often in the afternoon we get busier with, like, patients who come from surgery, because, like, towards the end of the day, then, like, there'll be post operative patients, yeah. So the they can either come from emergency, they might come as a book mission where, like, which is not as common, but like when I worked, my last one was orthopaedics, and they sometimes have like a book admission for like a hip replacement or a knee replacement or something, because they're not, like urgent surgeries. They're booked in to have surgery on the certain day, and they come in and then, yeah, anyway, so

they can yet be a booked admission. Or they can come from pacu or recovery, like from recovery, from theatres, and then, yeah, that's kind of it all. They can come from ICU sometimes as well, which is not as common, but yeah, they do as well, if they'd like, they might have been on the ward, and then they had to be transferred to ICU and come back.

**Tomas**: That's very interesting to me. There's a lot going on. I got a lot to, you know, a lot to ask about, if I wanted to, but you know, anyway, so what I wanted to find out. So you talked about observations for post operative patients. So one of the things I was curious with is sort of how much of that is manual and how much of that is automated. So what technologies are going on there where you're just kind of like you're being helped out by technology. And then what sort of things do you think you need to monitor to make sure that they're still, you know, working effectively?

**Nurse**: Yes, so that we have, like, automatic I don't know if you've ever been in the hospital. Have you ever seen them, but they're like these little machines on wheels, and they do like the blood pressure, and it's like what people would buy as well at home if they wanted to check their blood pressure, like, but, well, not the same thing. But, you know, it's all automatic. Like, it's automatic, we have a pulse ox, so that goes on your finger. It checks the heart rate and your oxygen saturation, like, how much your tissue oxygenation? And then we have the blood pressure cuff, which just pumps up automatically. Like, we don't have to do that. We just press a button and does it. And then temperature, because, if you're doing like, because when they come from theatres, like, post operatively, it's hourly for four hours, sometimes more often, but it's usually hourly for four hours. We have to check on women in like, four hourly after that, yeah, but it's very frequent. So if you would try, so what we sometimes, that is usually what we do. If we have a low blood pressure or a high blood pressure, usually low, more commonly, that is like, that? We're like, Oh, that's too low. Is that even correct? Because manual, we did learn manual at uni, because I'm, like, checking heart rate and everything manually as well. Because, like, it is still important when machines fail. So if it's low, we'll do a manual to check, like, is it actually accurate. And sometimes it'll just be above the mark, like it'll be off, it's final manual, or sometimes it'll actually be low. So we do that, I would say I use the manual blood pressure machine, probably, like once a week, maybe, maybe once a shift, like, maybe not every shift. It is a really useful tool because you actually, instead of just pressing a button, you're hearing the systolic and, like, you're hearing the vessel. So, yeah, yeah, it's not really relevant, but, like, it's accurate, but it's obviously time consuming, and I'm grateful for the automated stuff, but, um Yeah, so yeah, but I mean in terms of, um, those automated things, they're different that you would would you say that they save you quite a lot of time, and in their functioning generally,

**Tomas**: Yeah, quite relied upon, especially like those, these pulse oximeters and the vitals monitors, those are pretty much standardised now, you know, yeah, so, yeah, that's definitely I've seen that with it feels like it's a bit of an increasing trend just this automation, but that's been around for a while. The other thing I wanted to point out was about electronic health records, in particular, like documentation of proceedings. So I'm not sure of how much involvement your role has in terms of, like filing file documents. So this is my primary concern. My project is about how difficult and time consuming it is to just complete health records in the hospital and after, I mean, after it, I would like to find out about how technology could help with that. But I'd like to hear from you about, like, how you find that process in general?

**Nurse**: Yeah, I find it absolutely like, it's really, oh my god. It's so frustrating. It is because it feels like sometimes they, like, the hospital is like, they're always adding new things that you have to and it feels like they're like, what like, they're like, how much time do you want to spend on the computer? Like, I want to actually spend time with the patients? Actually, spend time with the patients, actually care for them, but no, I have to tick all these boxes. So, like a typical the amount of documentation, which the the minimum documentation you have to do in a shift is to write a note for your patient each shift. So, like, that is the bare minimum, because it just details, like what you've done for them and what happened in the shift. Because, yeah, if you don't mind that, yeah, it just Yeah. So that is really important. And you obviously have to put in there, like observations and chart that. And if they're getting like fluids, or you have to monitor how much they're weaning, you have to put that in, like, the chart as well. It just depends on how sick they are, and the sick they are, the more there's going to be to chart. But there's also stuff for people who aren't sick that are like, like, when someone's admitted, we have to do a risk screening pathway, which is asking questions like, how much drink do you smoke? Like, just say, General Admission questions. But it takes, it's very time consuming, and it's not something that I've always I'm like, not great at doing. I'm like, better, but like, yeah, it just takes a while. And, like, addition checklists, and then every time we get handover, we're supposed to do, like, a handover checklist. Like, oh, did you check that armband? Did you check the oxygen and suction at the bedside. Do you do all these things? And apparently that's mandatory. I don't always. I like, try and get to it, and I'm better with it now. But, like, the first place I worked, I was also, like, a baby nurse. Like, really, baby. And I didn't really, like, have I just, yeah, you just, it's just not a priority. Like, the priority is to look after your patients and leave on time. I didn't always leave on time. Obviously, I was just trying to get, like, the bare minimum done. But yeah, there is so much documentation. And like, when if someone has a fall, is not you have to do, like, update a bunch of things. You have to write like you have it just, there's all this, like, extra documentation. And then now they've introduced this thing called like, goals of care on the EMR, like, the electronic medical record, where you just have to put in like, even just like that, they're like, the most common thing is, like, with old people, is like that, there are falls tourists and there are pressure injury risk, because those are two of, like, the biggest contributors to them, like lengthening their hospital stay. So it's something that we want to avoid. And obviously, like, it's bad if they they sustain a full or, like a precious or in hospitals. Not right. So you were supposed to put in, like, the goals of care and interventions. But then we're supposed to do it for everyone, or that everyone's supposed to have goals of care. So they're just like, oh, just put those two in, like, the educators, like, Oh, just a tick. At the moment, it's just like a ticks box thing. It's something that they're trialling, and it's like, well, it just kind of defeats the point, because it's not like, we're not doing anything about that. Like, if they actually are, like, there's someone who can walk and they're not, they're moving around, they're not like they're and they're very stable on their feet, like they're not really at risk of either these things. But I still have to put it in. It's like, what, what am I doing this for? Whereas, like, sometimes it does make sense, but also, like, you're, I just doesn't feel like it's impacting actually, the way we are caring for them, like we're going to care for them the same way regardless. So there's just the amount of documentation is quite is just ridiculous. And yeah, I think the most frustrating thing is that it's not all in one place. It's all in different places. And it's like, well, can you just, like, I don't know, it would be really easy to just have it in one place, or, like, in just two places. And, yeah, I think, yeah, it's not smooth, like, the documentation of it, yeah, like, it could be worse, but I don't love that.

**Tomas**: No, fair enough. Absolutely, no, you will saying something about the sort of moving information; not necessarily moving information from one area to another, but, but if you're in a different area, you have to document in that area. And there's, there's like, is it difficult to like document in separate places? And you want to become have a combined sort of system, ideally to

make it…I think what I want to kind of touch on is, is the idea of, like, having sort of a standardised model of of documentation where, like, you know, with especially between hospitals, I'm curious, like, you get information from other sources, other health sources, and you have to include that and consider that when dealing with patients and sort of how the documents are different from each other in terms of, like, the way that they're filed, and how information, just generally, is moving around in the hospital. Do you think that there's sort of a bit of friction in terms of how you can transition information around the hospital?

**Nurse**: Yeah, like, it, like, within New South Wales Health, we all use like, the same system, so it's, it's okay between them, but like, you will have an even different if they're in a different health district, if they get, like, transferred from, like, I don't know, Coffs Harbour or something, to North Shore, then they, like, have a different number, like the MRN, like, the medical reference number is different, I think so. Like, there is, like, slight differences, but for the most part, the documentation is the same for the public health system. So that's good. I think for Northern Beaches, like it's when they're transferred. Some people are transferring from non beaches because they're just too unwell to manage there and or, like, they have some specific surgery that needs to be done. And they work a lot more with paper as well, like their whole ICU as well, as done with, like, these big paper sheets. So it definitely is in terms of transferring documents. I think they scan some things in that are important, but like, for the most part, those documents kind of stay Yeah. Like it's not Yeah, it's, I don't know if that answers your question so much, but…

**Tomas**: So it's not necessarily like a huge issue in terms of documents being too far apart in formatting or or things like that. Do those standards you're saying that the state has the same standards of documentation so it's reasonably consistent the information that you're getting in?

**Nurse**: Yeah? Like, you can kind of scroll through, like, back, yeah, to notes from, like, say, years ago about whether at a different facility, like, I'm pretty sure, yeah, it kind of does come through there. Yeah, it's just what I find hard to find is, like, if I'm looking for a patient's history, I'll often try and find, like, a note where they're admitted to emergency, whether it's listed in there. Like, it's really hard to just find some things. I'm like, where do I go for that? Yeah

**Tomas**: Okay, okay. Well, that's useful to hear, to be honest, the I guess the last thing, my last question I have. I don't want to do I don't want to spend too much, I don't want to use too much of your time. But the last thing I wanted to find out was the involvement, kind of, particularly of AI in some way. So sort of, I talked to another person before, about AI in particular, and and they were mentioning the fact that, in their belief, AI models were useful for more, sort of simple, sort of tedious tasks, because it can help a lot with those sort of things. And I'm curious, in your experience, what use does aI have in the hospital?

**Nurse**: I'm trying to think about like, what actually is like, what is classified as…

**Tomas**: Some examples, like, for instance, if you might get an automation of medical administration, like, how much medicine to administer for a patient, like, automatically going through formulas. Or you might have in an electronic health record you might have, like, automatic filling procedures, like, like it might do text filling, for instance, those sort of things like anything that

is a kind of a shortcut. I mean, it can be very simple. It can just be a simple form of technology. But I'm curious, like, if there's any sort of thing that is trying to reduce your workload in that way.

**Nurse**: I think there is like, yeah, there is text filling, like, if you're if they have a bag of fluids running, and but sometimes I feel like I just can't figure out how it works, and it's just, and like, often they might have a bag of fluids running, and then their blood pressure is low, and so the dog says, okay, run it double the right or something. And so then the bag that's running on the computer, like it's technically the right does not match, like, what is actually so like, when you click on, like, you kind of double click on the box for the six to seven o'clock of where you want to chat in that they've had 50 Mills an hour, or, like, 100 meals an hour. But no, it wants to do 50, because that's what's actually done. So like, it does help sometimes, like, if everything goes to plan, which it never does. And there is sometimes, like, on the MA like, the Medication Administration Record thing, they do have some time just trying to think. Usually, for the most part, we have to do the calculations because, like, it's sort of just like, like, they'll I'm not lazy kind of thing. Sometimes, oh, actually, no, that's incorrect. Because sometimes what on the computer, we have to do the calculations. But when we we have the pics. This is, I don't know if you've heard they like, where they come where the meds come out. And so a lot of the meds are in the pixels. And it's like, you just type in the patient's name, you scan your your code, and then you it opens up and it says, oh, remove. Like, if you only need two mils of the five mils of a medicine or something, it'll say, oh, yeah, this is how much you throw away. But those machines are also known like they always tell us, Oh, make sure you double check and obviously to your own calculation, because they're known to like, not always be correct. Usually with the calculations, they're fine. Oftentimes, the issue with those is that, because the patient's medication lists pop up on their patient, like, on their name. But sometimes they don't. Sometimes they're greyed out because, oh, it's like, because it's a half tablet, and they're like, oh, that it doesn't, like, compute with it. So you have to override, override them. Like, look up medication. So it's like, it is helpful, because you can, sometimes you can just click through and be like, Okay, those are the tablets I need. I don't have to look each one up. They actually changed that recently, like, sometime last year, maybe half with you last year. Or actually, no, like, yeah, like, about this time last year. They changed that from before, where you just look up what they had and like, Yeah, I do like it, because then hopefully you can just click on it, yeah. So that's, that's one way, I guess I'm trying to think there is also, like, where we can just, like, kind of do, like, duplicate results and stuff and copy, like, that's just, like, your regular copy and paste.

**Tomas**: So nothing too advanced, necessarily?

**Nurse**: not that I personally, no.

**Tomas**: I can be, well, that's, that's good. That’s kind of good for me. Have you heard a lot of people like, have you had a lot of people that are like, well, in my experience, like, there's a lot of stuff that's kind of in a bit of an infancy stage, but not necessarily implemented. You know, it's very hard to implement something and sort of prove that it's going to work. It's especially in a hospital setting. It's like you got to make sure it's working before you put it in. So I feel like it's very much like sceptical with AI tools in particular. So No, but that's definitely it kind of correlates with what I was expecting, sort of quite safe, you know, not necessarily to, like, they're not doing much of their own thinking, you know, that kind of thing. It just helps with, like, baseline phases, like baseline patients, maybe where they'd go screen for regular, you know, yeah, expected situations.

**Nurse**: This one, I'm not sure if this is considered AI, but like one that is actually really, I think it's more like it's not really AI, because it's, I don't know if it's considered, because it's a system that I can't remember what year it would have been, like 10 or 20. I think about 20 years ago, something happened with a girl that died in hospital, and it was because she had her observations were concerning, but it wasn't getting picked up like they were just sort of quite borderline, but they were still like out of range of what we would so they introduced this system called like, between the flags where they have red and yellow it's like, like, red and yellow flags, and you have a yellow zone and a red zone. So like, if your patient is in a yellow zone, you have to, or you're recommended to, based on your clinical judgement, to call, like, the doctor to call a clinical review, where they have to come within 30 minutes and see the patient. They always do, depending on the situation, but yeah, and then if they're in the red zone, it's usually means that they're quite like Concerningly unwell, and you have to call a rapid response, or they have to be there within 15 minutes, so that. So the way it works in the computer is that we put in the observations that we type them in, save them, and then it pops up with this, like, yellow alert, like, call a clinical review, or like, call a rapid response, big yellow or red banner. So it's like, triggering us to like, be like, Oh, okay, what have I or like, it also triggers you if you've accidentally entered something wrong, and you're like, oh, wait, no, that oxygen was not nine, it was 95 so, yeah, like it, yeah, Daddy's help. I don't know if it's considered AI, because that's kind of like a system that they introduced a lot.

**Tomas**: Yeah, it's kind of, it's a little bit intelligent, but it's maybe not AI, but it is very applicable. It's kind of similar to the way that AI would get involved, just sort of doing bumps and recommendations and those sorts of things. But, but, yeah, um, no, that's, that's very helpful. Um, I feel like I've sort of used up double the time that I said I was going to do.

**Nurse**: Oh, really, I didn't. Oh, you didn't say, that's fine.

**Tomas**: But, yeah, I'm really happy with this. I feel like I've kind of, you know, asked about, asked about, just what's going on in the hospitals, and then we got a bit on health records, a bit on, like, just general technologies. You know, it seems like, yeah, it seems like we covered a couple bases. So I'm pretty happy. So thank you. I can send you, like, once I done my final report, I can send you some sort of, like, findings or something that that show, like, what sort of, you know, AI system that I've decided on,

**Nurse**: Yeah, one thing I didn't mention with documentation that it's, again, it's not, it's kind of like an inbuilt computer system. It's not really AI, but yeah, we often do it with our notes. Sometimes I hate it and sometimes I love it, is that we create a template for ourselves, for our our documentation, which says like observation, or like airway, breathing, circulation, like, we kind of create like headings, and we just kind of fill it out, and it triggers, like, it makes sure we don't miss anything. But sometimes I find it like, it just restricts the way I write, and I prefer to sometimes just, like, write freely and just be like, Okay, this is what happened, kind of thing, yeah, I don't know depends on one minute, but that, I don't know if that's an AI thing.

**Tomas**: There's a little bit AI, yeah, it is actually so…

**Nurse**: Like, then when I type in like, note, it like, comes up with my with my template, or whatever, yeah.

**Tomas**: Yeah, right, yeah, no, those kinds of things are exactly what I'm talking about. Though it's really useful, yeah, I'm gonna hopefully do a bit of a bit more research on that, and then hopefully I find out something interesting.

**Interview 3 - Junior Doctor**

Tomas: Yeah so the first thing I would like to ask is about sort of the general rush of patience around a hospital so like I'm curious about how a uh how you might prioritize uh patients and how maybe technology can assist with that prioritization.

Doctor: Oh I talk to my experiences um as a medical student, it's inventcent in the ED um essentially the patient flow when someone comes in at a EED or brought in by ambulance, they'll be triaged based on a one to four category system. So I don't know if you're aware of that.

Tomas: No, I haven't really heard about that. Did prioritization, no.

Doctor: Yeah, so at least at Vinney's that's the way it happened, um where there were a sign of category one to four, uh number one being US seen immediately by um, you know, by, you know, a senior doctor, the registrar or consultant. Cat two meaning you were seen within five minutes on arrival at three, uh I think it is you were seen within half an hour and cat four is maybe seen within two hours. I think they have specific targets. um that they try to meet based on what category they fall into. When they come in, um the triage nurse, so if they presented themselves and they just walked into the hospital, they'll be sat down in the bay and then the triage nurse will come by and assign them a category on a very limited um kind of history and then, you know, based on what category they are they'll be seen in a certain amount of time. Okay. a doctor's end, we have a view of basically all the patients that are sitting there and we have a look um we kind of see each patient um, according to their priorities. So some some doctors are assigned like lower priority um patients and that might be a f fast track or short stay unit. um versus the main adults, ED, which is usually the more unwell patients that need to be seen U um and there'll be, you know, essentially prioritized on our end.

Tomas: Okay okay um the way it worked well, I guess the technology-wise, um that's kind of the, yeah, like are you looking for solutions to improve that process or are you looking for um you know, how it actually works? Well, in terms of like, like you were mentioning about the triage system and the nurse kind of it's a very brief assessment of of the sort of urgency. So so I'm wondering like, I mean, is it kind of a screening based on just the just how they present or is there any sort of tools that maybe like maybe any documentation goes into it as to like their previous medical history or or how how kind of automated is that process? Or is it just kind of person by person?

Doctor:Basically, everyone else will have a set of odds, so just standard odds um taken there and which tells us like, okay, if your os are unstable, then you need immediate attention. um and then the triage nurse has specific training, um, which allows them to, you know, triage people according to, you know, how unwell they are. So they'll ask, you know, if anyone it's kind of protocol driven from my understanding as a nurse, um So I just don't know if there's an element of automation to um data assessment.

Tomas: Okay. That's fair

Doctor: But yeah, I guess it could be like the the transists will record a brief summary like presenting with chestain background of cardiovascular factors that should raise alarm bells and they'll be seen earlier. So, you know, you kind of like there's a an option to automate that element of triage. However, it's ultimately our nurse's responsibility to assign a category.

Tomas: Okay. Perfect, that makes a lot of sense. Yeah, yeah. um so the other thing I wanted to find out uh to do with that is kind of like more particularly when you when a patient comes through and is being dealt with by a doctor um and sort of the tools that might sort of automate their a patient's care and maybe um so for instance you might have to conserve your energy for what for a more manual task and maybe you don't have to worry too much about taking their blood pressure or so on and so forth like uh are there any particular sorts of technologies which um sort of help you in that process of of taking a patient's well dealing with patient care in particular in an ED setting

Doctor: what would um, you know, useful is to have like an AI scribe that would be amazing. I just don't think um, you know, it's been approved because of the cost and because of the potential um legal consequences of like an AI scribe getting things wrong um but that, you know, if you ordinate documentation based on, you know, and either a camera or looking at the console or a, you know, a microphone recording what was said, um Ross, that would be a huge time saver. Yeahment also automation if you um I guess like there are alerts for when os do get deranged and someone needs to physically do those os so that will be a nurse that goes around on a, you know, like with a certain frequency, having os every, you know, 15 minutes for the unwell patients or every hour or two for the, you know, non-unwell patients. um So I just don't know if there's an element of automation on that level, but there could be a better alert system um as to when I start declining.

Tomas: Sure, okay. And in terms of diagnostics in particular, where do you do you feel like there's a limit where maybe AI tools uh can sort of extend their domain in terms of the safety of of like, of course, these um these scribes aren't even approved, right? Because of the legal consequences. So but in terms of like uh realistic safety of an AI tool, do you think that diagnostics would be kind of compromising it would be maybe a bit more questionable for lack of a better word.

Doctor: I think it depends on the applications, um like a simple chestray, which has been done knowing to the times over and it's a standardized process. I'm sure there's AI technology that actually performs better than a consultant radio radiologist. Like a door there's there's there's things that AI can tell you based on weird metrics about the patient's race, you know, from a rate, like just weird things. So all there's a benefit to AI, you know, looking at a scan and a what they think is wrong with the scan, but it's ultimately you need a you need someone medical to rationalize whether those results aligned with what's happening clinically in like to the patient. So they might say gallbladders, you know, contracted and it looks inflamed, but that might not align with what's happening to the patient there and then. So I don't think I'll ever replace the clinician, but it will definitely assist. And I think, you know, future, you know, in the future there's going to be less radial radiologists because you know AI do it better, but also you'll also need someone at their upper end to confirm the outlieses that aren't easily identified.

Tomas: Sure, yeah, okay. So um in in maybe a broad sense uh and I won't ask or maybe ask just this question so we can you know I don't want to take up too much of your time but um I guess the last thing that I wanted to find out was just as an AI tool then, do you think it's uh do you think it is more applicable for a tool to be sort of helping with more mundane processes which are kind of

time consuming but not potentially not potentially like risky? uh Do you think that's a better setting for AI tools to be implemented and, you know, in terms of the safety of their implementation?

Doctor: Absolutely. I'm all for um AI improving the efficiency of patient care and patient safety. But you know, it just depends on each use and it's like application. uh What are the pros and cons of each thing and each particular context. So you know it's it's a hard question to say overall, but absolutely.

Tomas: Yeah, that makes sense. I mean in terms of um so is it sort of crucial that uh that there's sort of an oversight element like when you go into diagnostics it's it's maintained that it would be important for a doctor to sort of cross-check because sometimes sometimes you hear things about AI tools uh inducing some complacency or or like a a less like doctors might take a step back and rely on an AI tool, but it's probably quite important that you don't do such a thing as you allow it to save time where it can. But then when it comes to the uh the diagnosis as a whole, it still maintained that it's that it's the clinician's role to to find an accurate diagnosis.

Doctor: I think that's with most things to be honest. I think I think doctors and like the medical practice in general is very data driven. So if you can run a study that proves that um this AI tool performed better than a consultant radiologist and have better, you know, uh, you know, accuracy then I don't see why that's a barriot implementing that particular tool. um The problem is it's just training. I think it's the AI that's not there yet because there's such a vast amount of data um and like that need to be fed such a vast amount of like um scans to accurately pick up every case. Yeah. um I think for for routine cases, it's fantastic, but for anything more complex, it should be flagged to have a you know a consultant radiologist look out it um I think I forgot your your initial question.

Tomas: Oh no, no, absolutely there was just just sort of about like um maintaining uh an oversight when it's down to that instance uh, where there's sort of consequence on the diagnosis. You know, the importance of maintain of a doctor being there to check on the AI tool and for it not to be relied on. you know, if you can prove the AI tools is better than the consultant radiologist, shouldn't we be checking on the consultant radiologist?

Doctor: Yess absolutely true. It's a very good everything will have its you know sensitivity and specificity and things that will fall through the cracks. But you just got to prove, you know, it happens to a lesser degree with an AI tool. And I think that can be more wisely accepted. And if it has efficiency gains, which it will or have costs saving or have more efficient diagnoses, you can scan someone at an immediately know a diagnosis That would be fantastic. But, you know, it's just whether the health system is um you know, accepting of the new technology given, you know, how it performs against the current practice.

Tomas: Sure. Yeah. That makes a lot of sense. um Yeah, look, I'm I'm pretty happy with that. I'm not sure if you were, uh, you know, I heard that you're bit in a bit of a rush so I don't want to kind of press you too much for time. Okay, okay well, yeah, I mean look, I'm I'm very curious about all these things and, you know, kind of I mean suggestions of where AI could help or are you just talking about general, how you know, how generally fits in? Well, yeah, I mean, I'm it's sort of both it's like, uh, I guess in terms of improvement, uh, I'm looking a little bit at at documentation to be specific in the moment. And um and you mentioned a scribe, which is something that I am actually quite fascinated about and and considering as something that could help healthcare. But um I was also looking at like like uh standardized formatting, you know, there's a there's a couple of other things. I talked to someone that mentioned that that the New South Wales health system has a

standardised format for documentation and they didn't necessarily believe that that documents were hard to integrate together. And I'm not sure if that's a consistent belief, maybe with you. Like what was that second part like to sugar out together? What In terms of like if you if youive patient medical records from different sources it's so hard to add that to the file.

Doctor: Yeah. So yeah, so I guess there's two things. It's like I guess two things that I want to say. um is I'm all for an AI scribe. They pick up what's going on in the room, but they don't pick up what's going on in your head. And of and there are things that are very different going on in your head when you're dealing with patients who are like delirious and confused and, you know, dealing with families and distress, you're not going to say certain things um for the sake of a scribe. If it's going to upset the patient. That's when, okay, yeah, it could help. you know, and and do a preform added note for you and, you know, highlight everything that was done in the consultation, but you still have to go in and input your exam findings and input your impression and your plan. um But for the first you know, the most of it, it will be an asset and I'wise not already done. Sure. Yeah. um I guess the second part of that was uh about the as of uploading sort of different sources of information into the one medical record. um Essentially there's I think there's a lot of hist, you know differences and difficulties between local health districts that have different privacy laws. So there's different systems that allow you to access patient confidential information. um that, you know, for a user that's been granted to that uh information within the local health district for the purposes of caring for that patient. um So it's kind of everything's under lock and key and it's kind of hard to get, you know, specialists letters that are, you know, done in private practice. It's hard to integrate that into the medical record unless you're like manually transcribe it or upload it via pedDF um So you know, some, some just like annoying difficulties there. It's not impossible, but it's just annoying yeah some some tedious sort of some barriers maybe.

Tomas: Yeah, totally. Okay. That makes sense. Well, I mean those kind of those kind of gaps I feel like from my research it feels like those kind of gaps are where AI can be pretty particularly helpful in sort of preformatting and and, you know, aligning documents so that it's just easy to integrate. Those kind of things could um have a bit of a gap uh to assist the healthcare, but yeah I think it's a huge difference.

Doctor: I don't I think like the AI scribe would be uh like game changer and like AI preformatting a ward round note would be a game changer. um if you just like the the other document Iload would be like a nice little assistant uh like to maybe have like some time saving but you know I I might every maybe once a week or once or twice a week need to upload a specialist letter to a patient file um you know and I I guess you could have an AI to look over that file and convert it from a PDF image to text if you need to. But you know, I don't know, you know, if that would be a huge asset or not. Okay, that makes us a little. Yeah, I mean, um the you know what would be having the knowledge that this person went to a specialist on this date and this was the result ofentimes I patients come in confused. You don't know which specialists they are under and if they have like recent letters or, you know, GP consults um and you don't know any of that unless you specifically ask and chase down those letters, which almost no one does. Yeah. And and and how would that work in terms of those those documents are online and available, are they? No, necessarily. So, you know, if I take a GP practice, H GP practice we have locally stored files on their own system that won't be shared with um a hospital. The only opportunity they have to share with a hospital is if they upload a particular document to healthy, which is that O's system um for patients to have like a universal kind of like medical record. So it's kind of there's another barrier there that the GPS to physically upload everything to that server and they don't have time in their 15 minute console to do any of that. um and also you know like specialists won't do that either um because they're often

too busy but you know it would be nice to have an AI to automate that process and have information available at the ready

Tomas: Yeah, that's a good point. um okay look, I'll I'll maybe just finish off with uh with a bit of a question for you. um And because I've asked you quite quite particular questions, I feel like I've led you down some paths, but so, you know, if I just wanted to ask you like, do you think there's any time or automation that that could be introduced that maybe we haven't talked about yet?

Doctor: I guess, um this is maybe like a risk management system. um for it could be done in a ward-based or an ED setting where an AI scribe might look through a patient's progress notes and assign it a level of risk So, you know, oftentimes, you know, when a patient deteriorates will get a call and it will be a clinical review for blood pressure, you know, less than it should be. And like to us, that means nothing. It could be the patient's just a little bit dehydrated and that blood pressureures usually kind of low there an old grammar and they're fine. um or it could be someone who's significantly hypop perfused and dying. So to have a something to be able to scan the notes and actually come up with a more nuanced um, you know, risk assessment, then than just the os would be an asset because sometimes you've got multiple calls for patients, os that derange and like you don't know immediately who's the most sick, you can't have to call the nurse and ask what the story is and kind of give your, you know, give a quick assessment to decide who to go to first. So if there was a systemic way of like um integrating the os and having a AI run um, the, you know, an analysis of the of the progress notes to understand the patient's story and understand what's happening medically, then there'd be able to infer what's happening based, you know, with the os and assign at a level of risk that would it oruss about who to see first? Okay. That's like after hours, you know, when most of the hospital goes home at night and you're you know, like one doctor looking after a hundred patients you know what happens yeah definitely yeah.

Tomas: Okay. um I mean yeah that that's that's super informative. I definitely haven't considered a couple of those things that you've been talking about so I just definitely. idea is like,

Doctor: um you know, it's hard to get an idea of which teams in the hospital are busy or not. You you might have the same amount of patience, but you you don't know how complex each patient is and how many how much tasks are required for each So if you had an AI to look through um, you know, understand medically what's going on and understand the task that need to be performed for each patient, you could infer the amount of workload that the team is under um and then assign, um, you know, which team is, you know, understaffed or overstaffed and that balance out some of your rostering problems. Yeah, that's actually a good point as well about, you know, long work hours and, you know, I just I'm off an 11 hour day when my colleagues are going home on time consistently you know that would help admin who often have a very poor understanding of our workload. There's some some sort of like communication barriers there. I would I would gather because you guys wouldn't necessarily have the time to just chat about those sorts of things. No, no, like we don't we don't discuss our workload. We just work late. Exactly. You know, there's cardiologists out they're working till 10 at night. I've been working like 10 11 hour days. Pretty consistently um and you know, I might I it just happens to be I'm on a team of, you know, three people where there's other teams of four people and they seem to do better, you know, in a way of overseeing that uh from a rosterering perspective Yeah. um Well, they don't oversee it well is what I'm saying

Tomas: No, absolutely. That seems very difficult. So, no, those are there's definitely some there's definitely a lot of opportunities. I mean, you know, uh I feel like I feel like that's definitely covered off more than enough information for me, you know? I've got a lot..

**Interview 4: Receptionist at NBH**

**Tomas**: Okay, sweet, um, yeah, it shouldn't be too long. It should just be like 15, like 20 minutes, maybe max, and yeah, like, you don't have to - you can talk particularly or you can talk generally. The last person I talked to just wanted to speak generally because of confidentiality stuff. You can always just ask me to cut something if you want me to. So usually I start with a pretty general question, just to sort of get things rolling. So I might ask you, like, what would be, what is usually, your operations in your job, generally?

**Receptionist**: Okay, so I work within Northern Beaches Hospital. I'm a receptionist, and I'm part of the imaging department, so we have kind of like two main sections. We have an outpatient area and then an ED section. So outpatients is obviously people who aren't admitted to the hospital. Same goes for ED, they are outpatients, but then a lot of them go on to be admitted to the wards based on what imaging finds, what the doctors in the emergency department find, and their condition. So we're kind of a first step for a lot of what goes on in the emergency department. Doctors send patients for initial tests. I pretty much just book them in on the computer. When they come around to imaging, I arrive them, and then make sure that it's going to like the right doctors, that it's been booked in payment, whether they're international people, if they're Australian, if they have Medicare, all that kind of stuff. Because we do public, public and private within the hospital, obviously. And then I, when I'm in the emergency department, I'm the only receptionist. The radiographers take the images and then we send them back to the emergency department.

**Tomas**: Cool. So in, so in the outpatient area is that a different like, physical space to the emergency department area?

**Receptionist**: Yeah. So in the outpatient area, you're not the receptionist. There's usually five of us on the front and three out the back.

**Tomas**: Is that like, Would you say that's because there's a higher workload, generally?

**Receptionist**: Definitely. Yeah, we get a lot more people coming through outpatients, but also, just as in the ED, even if it is really busy, there's only so much that you can do there as a receptionist, it's pretty much just like getting the orders from the doctors and putting them into the system, whereas out the front we actually see the patients face to face, and we book them in for the scans differently than when the doctors kind of do most of that in the emergency department. So there's less to do in the ED than when we're working fully outpatients.

**Tomas**: Okay, cool. Um, so in both is like a general question for those two areas. Do you do like patient prioritisation? Is that left up to the doctors? Or if you book a patient in, are they kind of met with any sort of urgency depending on their situation?

**Receptionist**: Definitely. So you kind of have to have an understanding of certain conditions, even just as a receptionist, because, well, doctors will write urgent. If we get a referral, a doctor's written urgent on it. In either case, we scan it into the system, one of us will go and tell a radiographer or tell our manager, hey, this needs to be done now. And we also have something in our programme where we can change the urgency. So if a scan has been done and the patient presented normally, but The radiographer notices there's something huge going on, then we can go in and change it to urgent reporting, and one of the radiologists will pick it up straight away, yeah. Same goes for the emergency department, because we print out all the referrals, pass them along to the wardsman, who will go grab the patient. If we see something that is like a stroke or something like that, we

know the protocol is like, this needs to be done straight away. We will go tell one of the radiographers, who will, like, make them get the patient straight away.

**Tomas**: Okay, cool. So, like, the urgency in an emergency department, is that somewhat defined by the doctor before it comes to you?

**Receptionist**: Yeah, yeah.

**Tomas**: Right. And then the outpatient, because you're seeing the patient, is it more like you're the first line of calling that?

**Receptionist**: Yeah, yeah, I'd say so without patients, unless, like I said, the doctor's gone ahead and explicitly written urgent, yeah, it comes through us, and then mainly the radio radiographers, once they've got their list, will determine which order to do it in. But if I pick up something that I know is like, this scan needs to be done the same day, then we'll make it happen.

**Tomas**: Okay, sounds good. Um, okay. So in terms of documentation in the emergency department, are you mostly receiving requests for imaging bookings, that is, would you say that's the bulk of your…

**Receptionist**: Yeah. So essentially we yeah, we receive requests that we put straight in. So emergency department, generally, all the X rays are done straight away. All the CTs are done straight away. It's a bit confusing. So we have an x ray scanner and a CT scanner in the area that we're actually in in the ED and then we also have all the scanners in the outpatient section. Ultrasound is done over an outpatient. But we book emergency patients when we have enough spots, we often run out of spots, so that's something that we take for future, and we say, Oh, this might have to be done as an outpatient, or this might have to be done tomorrow, if they're still in the department tomorrow. And then same with MRIs. We do the same thing if there's space on the list for that day, the MRI team organised it so we ultrasound, X ray and CT. But then there are other scans that are ordered that have to kind of be triaged about whether this is something that's done urgently or this is something that can be done later as an outpatient, but yeah, we get referrals for every patient from the document.

**Tomas**: Cool, yeah. Um, and then in terms of, like, after a scan is conducted, is there anything that kind of comes back to you in terms of documentation, like, do you have to message the patients in any cases or send any information out afterwards?

**Receptionist**: Usually, no.

**Tomas**: So what happens when a scan is completed?

**Receptionist**: We book the scan in, we scan the referral into their file, and then the radiographers will gain consent beforehand. All the paperwork goes off to them afterwards, so they also scan things into our system. They'll re scan the referral once it's been done, and they'll regain like the finished consent forms and everything. So it all does go into our system. We very rarely, will need to contact the patient after the scans been done. Sometimes, if it's incomplete, a radiographer will miss something and say, Oh, they can come back for like, another view. We'll call we'll recall them. But generally speaking, that we send an automated message about accessing results, but we don't really have anything,

**Tomas**: Yeah, that makes sense. So the system that you guys have, I'm not sure like to what extent you use each feature of the system, but do you have kind of a, is there anything you could say about kind of how it is laid out, like, where the different like are things easy to sort of find within the system in terms of documents and different things that you might need from it.

**Receptionist**: I think the way that the system set up is each of our imaging locations around New South Wales, you select the location that you're at, so we also have access to the other locations locally, which is helpful, because sometimes patients get some scans done at one and some of the other, and sometimes we need to book patients somewhere else When we don't have room. But regardless. So yeah, the system you pick your location, and then it has each modality set up, so x ray, CT, ultrasound, the doctors, all that kind of stuff. And we essentially can either pick a room and book a patient into that room, or we can go into the patient's file and add a scan from there. So we can easily search for the patients. We can search the daily work list to quickly pull up someone who's just had a scan or who's having a scan that day. Once you go onto a patient's folio, you have all their details, like personal details, obviously, and then you have a section you can open up that shows all of their past reports and scans. So you can access, yeah, the doctors can access their past scans for comparison, or that kind of thing. We can see if they've already had their scan done, or if they need something too booked in. Yeah. So there's, like, all the patients have their own file, and then there's a general system as well.

**Tomas**: Perfect, yeah. And are those things by standards like, is it a standard to upload those images or sort of appointment details? Or is it just kind of like, is it always going to be accessible that a patient's been booked in at another imaging place? Or is it their decision to display that up to you guys?

**Receptionist**: So it's always accessible if we open if we're just looking on our work list. So if I'm looking at the Northern Beaches hospital work list, and I search a patient's name and they've booked at Brookvale, it doesn't show up straight away, not their file and I see upcoming appointments, it shows all their upcoming appointments at any location.

**Tomas**: Okay, that makes sense, yeah, and that's not just for it's not just for you, you guys, like, is it for the whole statewide set of imaging?

**Receptionist**: Yeah.

**Tomas**: Okay, cool. So that's pretty that's pretty convenient, then.

**Receptionist**: It is convenient, but I will say as well, like we have the option, this is so random, but if patients want, we can say, don't upload to their health record.

**Tomas**: Okay, so you have a bit of autonomy about that.

**Receptionist**: Yeah, yeah.

**Tomas**: Okay, cool. So the other thing I wanted to find out with, and this is more of an opinionated question for you, in terms of what you might do on a day to day basis, do you feel like any particular parts of your work is more automatable, like capable of being sort of conducted in a more efficient manner, potentially with assistance, or?

**Receptionist**: definitely, I'm just trying to think about this. I feel like we get the majority of, like, majority of referrals we get handed to us. There's not much we can really do to streamline that process. Like, if the patient brings something in, we just work out what it is and we book it in, but we have phone lines that just go crazy. And there is the option when patients call in to select one for x rays, like two, for ultrasounds, like three, for whatever. And then we can choose which kind of lines we're answering. So sometimes they'll Okay, today you do MRI bookings and you do the other ones on the phone. But the thing is that people don't know, like, people don't understand often, like, what their referrals are actually saying. So then the phone lines are a bit just like scrambled anyway. So I guess kind of it'd be great if and with emails as well. Same thing people just send through their referrals. It would be great if we could kind of have a way for things to be, like, organised into modalities easier, yeah, which I definitely think AI could have something to do with, like reading a referral and like identifying this is the scam. Like, even though we can do that as well, it would save a lot of time just kind of that organisation aspect,

**Tomas**: Yeah, yeah. What I, what I um, seem to find out generally, is that, you know, AI is, like, good for boring stuff, if that makes sense. It's not necessarily too hard to do, but it just takes time. And it's, you know, it's very efficient at those kinds of things. So I definitely kind of agree with you on those sorts of things. But, um, but yeah, like, if there are any tasks that are like you were mentioning the numbering system, which I actually like, called in, like last a week or two ago to get a scan, and I had to go through that thing. So I'm aware…

**Receptionist**: Our MRI system sucks. Yeah, it's just a general issue that, because it's it's so frustrating. Like, we need to see the referral. It makes sense that we need to see the referral. I don't know if patient thinks it makes sense, but like, we need to know who the doctor is. We need to know exactly what to scan because, again, like some people don't understand, even if you ask them exactly to read off the referral, they might read like words wrong because it's medical jargon, so we need to see the referral, but then the team needs to take time. They're scanning all day, so they don't have a lot of time. And then we need to take the time to call. But we've got constant patients, and our phone lines are so full, it's like a choice between, do we answer the patients who are actively calling in, or do we call the patients who are waiting? Yeah, so that's very frustrating,

**Tomas**: No, yeah, that's when I, when I called Brookie, it was like, yeah, it didn't necessarily, or actually, they would have needed to scan the referral as well. Is that right?

**Receptionist**: Like, they, yeah, yeah.

**Tomas**: So yeah, because I was like, I called in and I was like, Oh, well, I can't book quite yet, because I need to get everything checked through. But it all Yeah, it makes a lot of sense, like, why that would happen. But, um, that's like that phone line system is, you know, a small example of it but do you think there are any other kind of already automated parts of your job that that are trying to make things a bit more easy for you?

**Receptionist**: Definitely the confirmation messages that get sent out. But an issue with our system is that when you set up a patient's folio, you have to tick a box that says that they're willing to receive SMS communications. And a lot of the time when you're quickly putting someone detail, someone's details in, you just like forget. So a lot of the time, we'll have a full list of patients booked for the next day, and we have to go through and check that that box has been ticked anyway. So it takes double the time. And then if it hasn't been ticked, we have to manually send out a confirmation message. So something that should be like, very simple, like, make sure that all the texts are sent out tomorrow takes, like, half an hour anyway, and then a lot of the time with specific

scans that are a bit more involved, we have to call to make sure that the patient is aware of the preparation as well, because if they show up without prep, then we have to reschedule them. So that takes a good chunk out of our time as well. So I guess that is something that could potentially be automated. But then we also kind of need to know ourselves that they like, actually understand what we're telling them, especially with older patients.

**Tomas**: Yeah, definitely. So like these, these, like emails, the box ticking, like consent, things where they're quite like, if you in the process of taking it in the first place, are you verbally getting their agreement, or is it sort of like, what's the situation where you're supposed to get that confirmation from them?

**Receptionist**: Honestly, no, because when we take a pic, like, we give them a new patient form, where we ask them, like, make sure that you pop in here. I guess maybe some people might say, if you're putting your phone number down, or you're okay with us sending you a text. But I think nowadays might be like bad to say, but a lot of people, like, I get a million texts from all these appointments that I make, like, a lot of people kind of assume. And then you get people sometimes asking, like, straight away, can you just send me a message straight away? Blah, blah, blah. And then I go onto their account and I see that the box isn't checked. So it's something that a lot of people, I think, expect. And then we have that issue as well, when if we accidentally haven't ticked their thing, people will call up a few days before and say, I haven't gotten my reminder yet. And I'm like, Okay, so, like, we know that they are happy to get reminders, but in general, I think it's kind of like a take. If they're writing down their phone number, they kind of have an understanding that you may need to contact them for something. Same goes with email, because we take that as well.

**Tomas**: That makes sense. Okay, um, perfect. Yeah, I think we're pretty much on time. Now, I feel like, you know, I've I've got, this is, like my fourth interview. Now, yeah, pretty, you know, feeling a lot better than I did before my interviews. No, there's some cool stuff, um, the like, I didn't know you were kind of in the emergency department as well.

**Receptionist**: I feel like I didn't talk much about that. We also do have just quickly, we have, like, another programme that's specific to when you're in the ED. We have one that has all the patients that are currently in the emergency department, and yeah, so it shows their locations, because we need to access that, because we need to know where we're picking up patients from. Also, same goes for inpatients, because in the ED we organise some inpatient X rays, so we have that and then in that programme, all of the orders for the patients. This is not just an imaging thing, it's a hospital wide thing. So their main patient orders their the doctor when they come visit the patient. And we'll put notes on this website, so everything for the patient while their stay is in, the ed is there. So when an imaging request comes through, it comes through on this site, and then it gets sent over to our usual programme that we use, also for outpatients, like I've been talking about the whole time, but that link that's between them, it often goes down, and then doctors will have to write manual referrals and hand them over to us, like, we have planned outages, because it happens so often that we all need to know, like, what to do if either of them break down. Sometimes, if both of them go down, it's like a real disaster, because we have nothing online. So that's definitely an issue with using technology.

**Tomas**: That's a good point.

**Receptionist**: If, when both of them are down, it is like a free for all, because the new adopters don't know what's happening. So it hasn't happened to them before. We have people bringing in paper referrals and to print out the ones that have already been booked in for people. So it's a bit

of a bit of an issue when that happens, but then it is also, I guess, kind of a good thing, because it ensures that people know, like, the protocols in general, when you have to hand write it, like, we have doctors come in and I have to explain, make sure you put the patient's name on it and what bed they're in and, like, it's stuff that should be like, common sense, but we're so used to just Having it All automatically already in the system that people forget. So I think it can be a good reminder,

**Tomas**: What sort of situation would it go down in, just a power outage?

**Receptionist**: Yeah, power outage. I'm trying to remember. Sometimes the websites are just like, they literally just go, there's, they're just an outage. Like, yeah, it's quite common for the link between them to go out. I don't really know how it works. I just know that that's like, not sending the referral on, but both sites are still working, right? So that one's pretty easy. You just print it and then you put it in manually. Yeah, yeah. So most of the time that's what happened, is that they kind of just stopped communicating with each other. But then, I guess sometimes websites crash, like they get overloaded, yeah, definitely.

**Tomas**: But it could be, like, it could be something to do with the amount of information that's potentially been trying to pass through.

**Receptionist**: That's probably what it is, honestly, yeah, because most systems, like, there's a lot of stuff.